



A
Project Report
on
**Monument recognition from satellite images using deep
learning with explainable AI**

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by

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May, 2026

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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This is to certify that the Project Report entitled "Monument recognition from satellite images using deep learning with explainable AI", which is submitted by Sujal Gupta, Vansh Goel, Sumit Kumar and Samarth Saini in partial fulfilment of the requirement for the award of degree B. Tech. in Department of Computer Science & Engineering of Dr A.P.J. Abdul Kalam Technical University, Lucknow, is a record of the candidate's own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ABSTRACT

Recognizing monuments in satellite images is a challenging problem because of differences in spatial resolution, illumination conditions, season, and high-level clutter in the background. Moreover, monuments usually constitute a minor fraction of the whole image and have different scales and rotations that hinder their reliable recognition with conventional methods [10]. For addressing these problems, in this paper we introduce IndiSight++, a framework based on deep learning for efficient and interpretable monument recognition.

IndiSight++ uses MobileNetV1 [2] as the main component due to the use of depth wise separable convolutions to ensure computation efficiency and effectiveness in feature extraction. The preprocessing and data augmentation phases include normalization, resizing, rotations, scaling, and changes in brightness levels of the input images. Furthermore, to minimize background cluttering and direct model attention to relevant parts of the image structure, the saliency map guided cropping technique is used [8], [9]. In order to make the model decisions more interpretable, various explainable artificial intelligence (XAI) models are included, which include the Grad-CAM method [3], LIME [4], and SHAP values [5]. The proposed model is trained and tested on a custom dataset including 10 categories of monuments with satellite images, reaching a test accuracy level of 92.7%. Results obtained from experiments confirm that the methodology presented not only produces high classification accuracy but also guarantees interpretation by emphasizing features associated specifically with monuments. Thus, IndiSight++ can be considered a dependable method to apply to projects dealing with cultural heritage preservation, urban planning, and GIS studies.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
XAI	Explainable Artificial Intelligence
Grad-CAM	Gradient-weighted Class Activation Mapping
LIME	Local Interpretable Model-Agnostic Explanations
SHAP	Shapley Additive Explanations
MobileNetV1	Mobile Network Version 1
ReLU	Rectified Linear Unit
SGD	Stochastic Gradient Descent
Adam	Adaptive Moment Estimation
SIFT	Scale-Invariant Feature Transform
SURF	Speeded-Up Robust Features
BoVW	Bag of Visual Words
SVM	Support Vector Machine
k-NN	k-Nearest Neighbors
GIS	Geographic Information System
ISRO	Indian Space Research Organization
ImageNet	Large Scale Image Dataset
FLOPs	Floating Point Operations

CHAPTER 1

INTRODUCTION

1.1 Introduction

The architecture of India is well-known throughout the world because it has an abundance of buildings that have been constructed over many millennia by the efforts of several dynasties and civilizations. Buildings ranging from intricately carved temples to majestic Mughal forts and elegant royal palaces can be found in India, and they are all considered to be exceptional works of art, engineering, spirituality, and history. The Taj Mahal, Qutub Minar, and Hampi Monuments are examples of structures that are both architectural wonders and important pieces of India's cultural history. These buildings can provide insight into construction methods from ancient times, social and political structures, and cultural practices. As depicted in Figure 1, satellite pictures offer a bird's-eye view of these types of monuments.



Figure 1: Taj Mahal Satellite view

The significance of monuments goes much beyond just aesthetics and history. They hold great importance in many fields including heritage preservation, tourism promotion, education, archaeology, and even environmental monitoring. For example, preservation of monuments ensures the continuity of culture and heritage, whereas their use for tourism purposes has a major impact on the economic conditions of a country. Besides,

monuments often serve as an important subject of study in terms of architecture and history. For instance, they provide valuable information about building techniques used by ancient people. With regards to urban planning, knowledge of the spatial distribution of monuments enables city authorities to develop infrastructure that does not infringe upon the heritage zones. As evident from Figure 2, historical monuments like the Qutub Minar have immense architectural significance.



Figure 2: Qutab Minar

Thus, correct identification, tracking, and preservation of monuments are critical steps in achieving sustainable development and preservation of cultures. Conventionally, documentation and identification of ancient monuments used to be done manually with the help of surveying, excavating, inspecting by experts and documentation of historic records. Though these methodologies can produce satisfactory results, yet they are inherently time-taking, costly, and unscalable. The manual surveying of expansive and diverse geographic locations such as those in India can take an inordinate amount of time, money, and manpower to complete successfully. Further, some of the monuments are situated in remote locations that include dense forests, desert lands, or hilly regions that make their monitoring difficult.

However, thanks to advancements in technology, particularly satellite imaging

technology, and availability of satellite pictures online at places like Google Earth and Indian Space Research Organization (ISRO), it is now relatively easy to monitor and study expansive geographic areas without spending much effort and money. Satellite images are an effective and scalable way to conduct research on monuments and other structures as well as keep monitoring them for further changes. They can reveal potential dangers for monuments and even allow for discovering new unknown sites. In addition, satellite data may also be combined with GIS to offer a complete spatial analysis system [10]. As depicted in Fig. 3 below, there are several UNESCO World Heritage sites in India located in various locations.

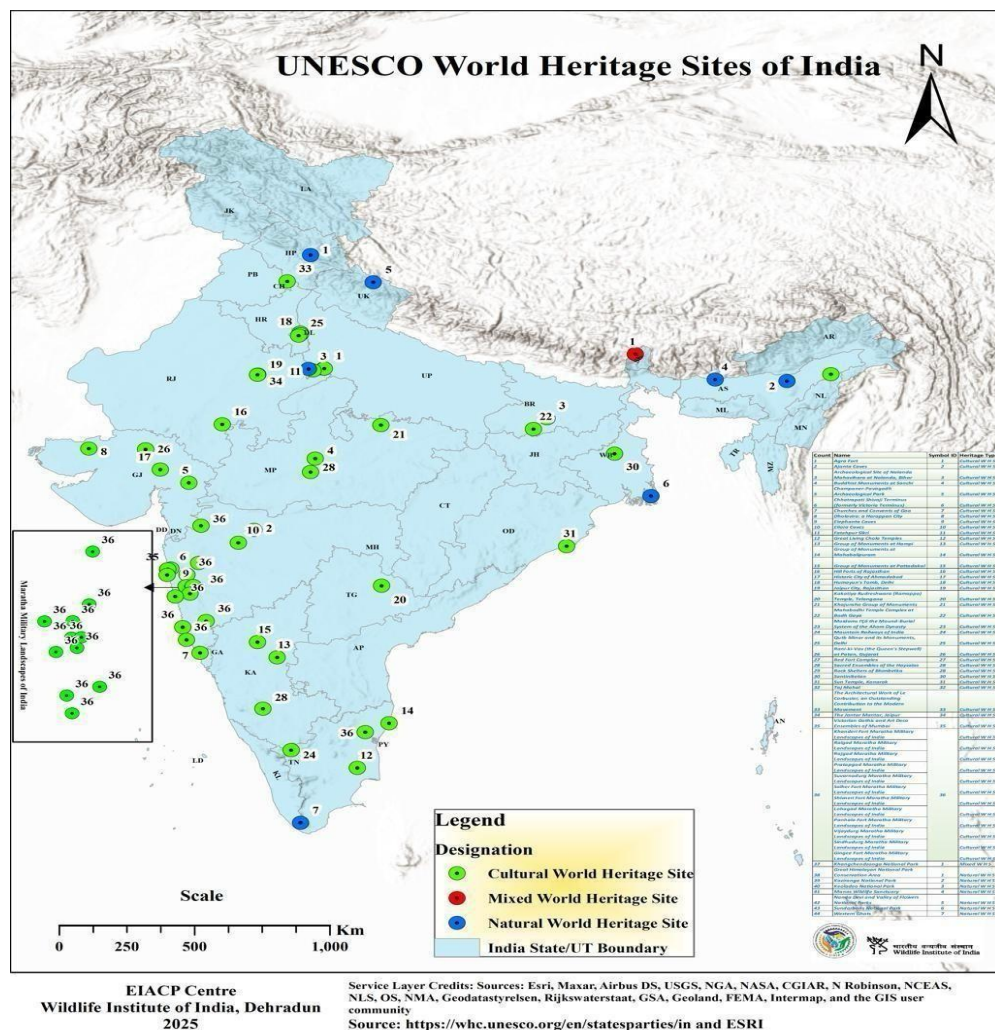


Figure 3: UNESCO World Heritage Sites of India

The introduction of the deep learning algorithms, especially Convolutional Neural

Networks (CNNs), in recent years has made revolutionary progress in the domain of computer vision. CNNs prove to be extremely efficient at automatic feature extraction from images, which makes them well-suited for the problems of image classification, object recognition, and semantic segmentation [1]. Algorithms such as MobileNet, ResNet, and VGGNet proved to be very successful in many domains [1], [2]. When used for satellite imagery, these models are capable of recognizing patterns related to monuments through their geometry, texture, and structure.

Nevertheless, although deep learning algorithms exhibit outstanding performance capabilities, they have one crucial disadvantage, namely, an absence of interpretability. Deep learning models tend to be called “black box models” because their decision-making mechanisms remain unknown and opaque. While the algorithm demonstrates excellent results, it remains unknown what leads to a certain decision. This issue becomes critical in applications that require trust, accountability, and accuracy.

One of the important issues in recognizing monuments with a deep learning model is to avoid learning from irrelevant information that does not add any value to recognizing certain classes. The satellite imagery itself can be complex and contain various entities like vegetation, road, building, water body, or even shadows. A wrongly trained model can mistakenly learn from these entities, associating their presence with some class of monuments rather than the actual structure of the monument. For instance, a deep learning model may use the surrounding greenery to recognize a monument, which can lead to poor generalization results when applied on a completely different dataset.

The idea of Explainable Artificial Intelligence (XAI) serves as an excellent remedy for these problems. XAI aims at increasing transparency and interpretability of deep learning models. The key aspect here is to explain what kind of input data is used when generating a particular output from a neural network. This helps understand what parts of the input image affect the final output and whether the machine learning model learned something useful and not something trivial. The widely known examples of XAI methods are Grad-CAM [3], LIME [4], and SHAP [5].

Grad-CAM produces heatmaps that indicate the parts of the image which are most

important to the model's prediction. LIME uses a local approximation of the model to explain each prediction by altering features and monitoring their impact on the result. Finally, SHAP applies a concept from game theory to determine the importance score for every feature, allowing a complete understanding of the model's behavior. Thus, these approaches form a complete solution for understanding deep neural networks.

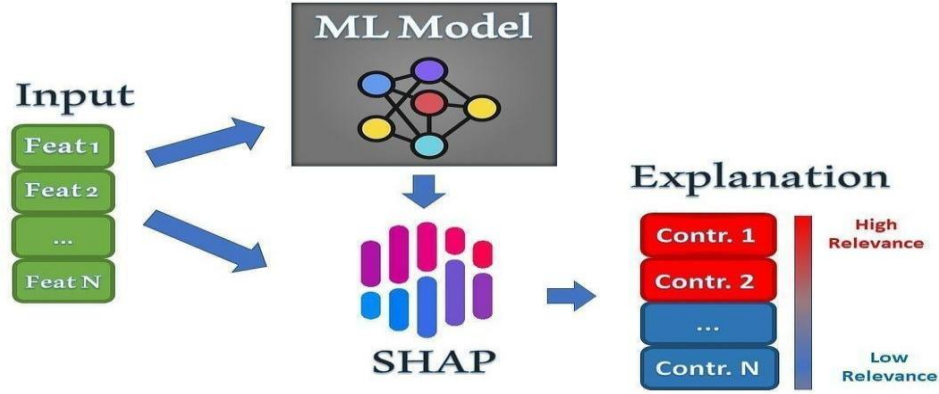


Figure 4: SHAP with ML

The significance of explainability becomes apparent when examining the case of satellite images because of the highly complex nature of the input data. Images captured by satellites may include several features at once along with different textures and multiple objects. Therefore, it becomes essential to verify if the model is identifying the correct aspects of the images. In this case, applying XAI tools allows for assessing whether the model is paying attention to the monument's architecture or some other unrelated features [12].

Explainable models can further help in debugging and improving performance of systems as well. Through analysis of visualization outputs, scientists will be able to detect situations in which models make wrong predictions and learn why it happened. Such information can then be used for improvement of data pre-processing techniques, model architecture, and training techniques as well. Moreover, through XAI it is possible to avoid biased predictions since such an approach does not allow relying on false or misleading features.

Despite its increasing significance, XAI has not yet been extensively used in remote

sensing and monument recognition area. Indeed, most modern models focus only on maximization of their performance without paying much attention to the way they work. It is evident that there is a clear demand for a combination of deep learning and explainable AI techniques.

Thus, this study proposes to develop such an integrated system for monument recognition which will allow obtaining results with high degree of accuracy. Through efficient use of CNNs and advanced preprocessing techniques, as well as explainable artificial intelligence, it will become possible to reach this goal. This system can be used to great advantage in many different applications including cultural heritage, remote sensing, and computer vision since it provides a robust means of recognizing monuments using satellite data.

1.2 Project Description

This study proposes a robust framework known as IndiSight++ which is designed for identifying, classifying, and interpreting the historical monuments from the satellite imagery. The framework is created by combining the benefits of advanced technology and methods related to deep learning, computer vision, and remote sensing techniques, leading to development of an effective, scalable, and highly accurate framework for recognizing the monuments through satellite imagery [10]. Instead of depending on the traditional methods that are primarily based on conducting surveys and inspections or use limited ground-based datasets, the proposed framework makes use of satellite imagery to monitor and analyze monuments.

The main objective of this framework includes two key aspects: first, accurately recognizing and classifying the monuments available in satellite imagery, and second, explaining and providing insights regarding the results provided by the model. While most of the frameworks only focus on optimizing their performance metrics such as classification accuracy, it is equally important to ensure that the framework is not only performant but also interpretable [12]. This is especially true when dealing with applications such as preservation of cultural heritage, geographic information systems, environment monitoring, or urban planning.

A key part of the proposed model architecture involves a light-weight deep learning algorithm known as MobileNet V1 [2]. The selection of this model is based on its efficiency and capacity to function effectively despite the lack of powerful computing capabilities. In this model, the depth wise separable convolution is utilized, an approach whereby the normal convolution is broken down into two parts, namely, depth-wise convolution and pointwise convolution. This makes it more efficient than the traditional convolution neural network since it uses less computation and fewer parameters. As such, the framework can process a large amount of satellite images without strain.

The dataset involved in this research comprises satellite images of different historical monuments obtained from public sources. The dataset comprises of diverse satellite images in terms of location, types of the historical monuments, scale, and environment. The dataset requires to go through a series of preprocessing tasks prior to using the model. Augmentation procedures, such as rotation, flipping, zooming, and brightness adjustment, are employed to enhance variability within the dataset and augment the generalization capability of the model when dealing with unseen data. Subsequently, the dataset is partitioned into training, validation, and testing sets to facilitate a balanced assessment of model performance.

The use of saliency-guided cropping is a strategy aimed at enhancing model performance by filtering out extraneous details in satellite imagery and focusing the attention of the neural network on architecturally important sections of the monuments. This is achieved through the process of isolating those salient patches in the images that incorporate relevant architectural components. As a result, any distracting artifacts, including vegetation and infrastructure, are eliminated from the images, allowing for enhanced prediction accuracy and decreased bias [8], [9]. In addition to this technique, the IndiSight++ architecture incorporates Explainable Artificial Intelligence (XAI) approaches, including Grad-CAM [3], LIME [4], and SHAP [5], which contribute to model interpretability.

The experimental findings prove that the developed IndiSight++ framework can recognize monuments accurately and efficiently. In particular, the use of explainable artificial intelligence (XAI) contributes to the better understanding of the model's

functioning and improves trust in the predictions. Furthermore, the visualization output proves that the network focuses mainly on structural and architectural attributes without taking external noise into account. Hence, the proposed approach confirms the effectiveness of saliency-driven preprocessing and explainability techniques implemented within the framework.

Furthermore, the work benefits from the rapid development of deep learning technologies, especially regarding convolutional neural networks (CNNs). CNNs have become a groundbreaking solution for automated feature extraction and hierarchical learning, allowing for significant advances in computer vision [1], [2]. The use of CNNs in remote sensing applications has provided an array of new possibilities, including land-use classification, environmental monitoring, disaster assessment, and infrastructure detection [13]. The proposed IndiSight++ framework represents another step in the evolution of CNNs, extending their application to monument recognition while overcoming associated interpretability limitations.

Consequently, the current research outlines a powerful, efficient, and interpretable architecture for recognizing monuments based on satellite images. Thanks to its innovative combination of lightweight deep learning approaches, modern image processing techniques, and explainable artificial intelligence tools, this system helps address the core challenges of precision, efficiency, and interpretability. At the same time, the developed IndiSight++ framework has a great chance to benefit various fields, such as remote sensing, preservation of cultural heritage, and intelligent geographic information systems.

First of all, Figure 5 demonstrates the fact that the satellite image shows multiple monuments and their surroundings, emphasizing the complexity of the real-life scenarios utilized during the experiments. It is important to note that the IndiSight++ framework itself is engineered with future implementation in mind. As stated above, satellite images are known to be characterized by such factors as resolution variance, lighting variations, seasonal influence, and atmosphere distortions, which might influence model performance negatively. In this regard, various preprocessing and augmentation techniques are used in the framework to counteract this problem [10].

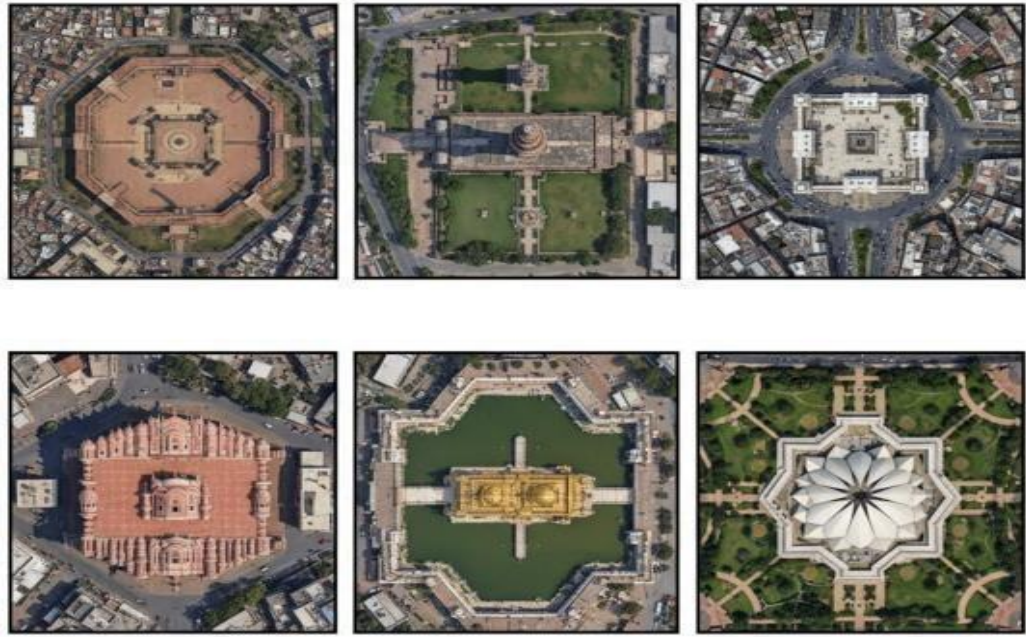


Figure 5: Satellite view of monuments

The next major advantage of the proposed solution is the efficient computations. Using MobileNetV1 provides higher speed of inference and lower computational costs, making the system applicable for extensive data processing and real-time application [2]. It is especially helpful in cases like environmental control and urban planning when immediate understanding of the situation is required.

Moreover, incorporating explainability into the process improves the convenience of using the solution for professionals in the field. Using explainable artificial intelligence methods allows validating the decision made by the model and eliminating any possible bias within it [12].

CHAPTER 2

LITERATURE REVIEW

Monument detection and recognition from images is a topic that has been receiving substantial consideration in computer vision, remote sensing, and artificial intelligence circles in recent years. Monuments and landmarks are cultural and historical treasures that signify the cultural and social values associated with a region. The recognition of these structures has proven important in applications such as monument conservation, development of tourist attractions, and digital preservation, among others. In light of increased demand for large scale analysis and monitoring, there has been a requirement for automatic solutions.

The growth and evolution of satellite imaging technology have made tremendous contributions in this field. Contemporary satellite technology has facilitated the generation of high-resolution images of the earth's surface. The rising availability of aerial/satellite images from public sources has also opened up many research opportunities in this field. Researchers can now develop intelligent techniques that will enable automatic detection, classification, and analysis of monuments and landmarks [10].

However, the methods utilized in monument recognition have gone through a lot of changes over time. Initially, the methods adopted for recognition included hand-crafted feature-based techniques where various features such as edge detection, texture analysis, shape analysis, and color histograms were hand-crafted from the given image. SIFT and SURF algorithms have been widely used in early image recognition models [6], [7]. Even though these methods performed reasonably well at certain levels, they had challenges when dealing with various types of distortions brought about by different lighting conditions, scale changes, orientation changes, and noisy backgrounds, leading to poor performance especially in real life applications.

Since then, there has been a tremendous change in the domain of image recognition with the introduction of deep learning techniques, especially the use of convolutional neural networks (CNN). Indeed, such networks have shown outstanding results when

performing the tasks of image classification, object detection, and scene understanding [1]. When applied in the field of monument recognition, the use of deep learning models allows achieving more accurate and reliable results in recognition tasks.

Despite the high efficiency and good quality of the results, such systems can hardly be called transparent. As it was mentioned before, these models can be called "black boxes", as they do not provide any explanation for their actions and decisions. It becomes rather problematic to build a certain level of trust between people and such a machine as a model may give misleading results.

In order to solve the problem, explainable artificial intelligence (XAI) methods have been created. The application of XAI allows explaining the inner logic of the model and visualizing its work. Using such methods as Grad-CAM [3], LIME [4], and SHAP [5], one can reveal which parts of the image were used by a model for making decisions, and whether these decisions have a sense and correspond to the reality. The utilization of XAI in remote sensing has resulted in the further enhancement of deep learning-based model interpretability [12].

In relation to the reviewed researches in this field, there could be identified three main directions. The first one is traditional feature-based approaches based on manually constructed features used in image analysis. The second category is deep learning-based techniques that use advanced neural network structures to discover intricate relationships within data. The last one comprises explainable AI approaches aimed at the interpretation and verification of predictions performed by deep learning systems. Each of these categories has specific strengths and weaknesses, and modern studies tend to combine deep learning with explainability in order to develop efficient and reliable systems [13].

To summarize, the incorporation of satellite imaging, deep learning, and explainable AI into monument detection applications creates new opportunities for developing automatic recognition systems. Not only does this method ensure higher precision of detection systems but also guarantees the interpretability of the obtained outcomes. With advancements in technologies, these intelligent systems are anticipated to make an important contribution in preserving cultural heritage and providing data-driven

decisions.

2.1 Traditional Feature-Based Methods

However, before deep learning-based approaches were developed, there existed another image recognition system, which employed manual feature extraction for achieving its goal. Traditional methods played a pivotal role in creating early computer vision systems and became the very first effort to teach machines how to analyze and interpret visual information. At their core, these methods implied the creation of an algorithm, which was supposed to detect informative and discriminative features within images. In turn, such features should have been used for describing various visual characteristics of objects or landmarks in an effective way.

In particular, one of the key elements in traditional image recognition approaches was feature extraction. This stage played a crucial role in terms of determining the overall success of the whole process since the quality of the extracted information affected the efficiency of all further operations. Specifically, researchers attempted to detect visual patterns, including edges, corners, textures, shapes, intensity gradients, and other similar attributes. Such patterns are often selected due to their high level of stability under various imaging conditions (i.e., illumination, scale, and orientation).

After extracting these hand-crafted features, they could be fed to classical machine learning classifiers, like SVM (Support Vector Machine), k-NN (k-Nearest Neighbours), Decision Tree and Random Forest. The aforementioned approach, where feature extraction preceded by feature classification, was dominant in computer vision for many years. It consisted of developing and fine-tuning each step separately, which required significant knowledge in the field. Nevertheless, this methodology showed considerable effectiveness and served as a basis for further developments in computer vision.

Probably the most important contribution into this area came from an algorithm called Scale-Invariant Feature Transform (SIFT). Proposed by David Lowe in 2004 [6], SIFT can be considered a revolution in feature extraction and description as it provides a way to detect and describe local features that are robust to changes in scale and rotation and resistant to changes in illumination and viewing angles.

In general, the operation of the SIFT algorithm is divided into several steps. First, the algorithm generates the image's scale-space representation by applying successive levels of blurring using Gaussian filters. Next, the Difference of Gaussian technique is used to locate the extrema of the image in the scale space, which correspond to the keypoints. Keypoints indicate areas of the image that remain invariant across various scales. After the keypoints are found, keypoint localization takes place, which improves their accuracy and removes unstable keypoints like those with low contrast or bad localization.

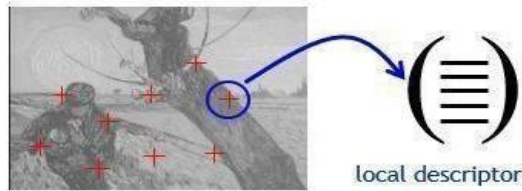
Once the keypoints are localized and verified, the SIFT algorithm sets an orientation for each keypoint that corresponds to the principal direction of the gradient in the neighbourhood of the keypoint. This makes keypoints rotation-invariant because it allows recognizing the same feature no matter what orientation it has in the image. Finally, the keypoint description takes place, where a vector of numbers is generated that describes the gradient orientations in the neighbourhood of the keypoint. Because these descriptors are distinctive enough, they can be used to match keypoints between various images and perform tasks like object recognition, image comparison, and panorama stitching, as seen in Figure 6.

Because of its robustness and reliability, the SIFT algorithm was rapidly adopted as a benchmark approach for computer vision. SIFT was commonly employed in various areas like object recognition, image stitching, 3D modeling, and landmark detection. The versatility of SIFT in handling changes in scale, orientation, and lighting made it an ideal choice for practical applications.

Following the success of SIFT, another critical development in hand-crafted feature detection is the Speeded-Up Robust Feature (SURF), created by Herbert Bay et al. in 2006 [7]. As its name suggests, SURF was created as an improved, faster, yet more computationally economical replacement for SIFT.

SIFT stages:

- | | |
|---|------------|
| • 1. Scale-space <u>extrema</u> detection | detector |
| • 2. Keypoint localization | |
| • 3. Orientation assignment | descriptor |
| • 4. Keypoint descriptor | |



A 500x500 image gives about 2000 features

Figure 6: SIFT STAGES

SURF becomes highly efficient owing to the usage of integral images that ensure the fast computation of box filters. Unlike Difference of Gaussian, which is utilized in detecting key-points, SURF uses the Hessian matrix-based detector, which enables efficient detection of blobs in the picture. Moreover, SURF makes use of reduced feature descriptors based on Haar wavelets that make computations easier.

Apart from developing algorithms of feature extraction, researchers also came up with the idea of using other types of information in order to improve the classification results. For instance, the Bag-of-Visual-Words (BoVW) technique is applied in image representation and classification, and this model is similar to the one that is commonly used in natural language processing. Here, the whole image can be considered as an ensemble of visual words, or local image features. The BoVW model is illustrated below.

The algorithm starts with the extraction of local features through techniques like SIFT and SURF. These local features are subsequently grouped using algorithms such as k-means clustering into a codebook. This codebook contains a number of “visual words” or clusters. The image representations are then computed from the codebook through histograms that represent the number of times the visual words occur in the image. Using this fixed size vector, classification can now be done using any classical machine learning classifier, such as SVM, k-NN, or Random Forests.

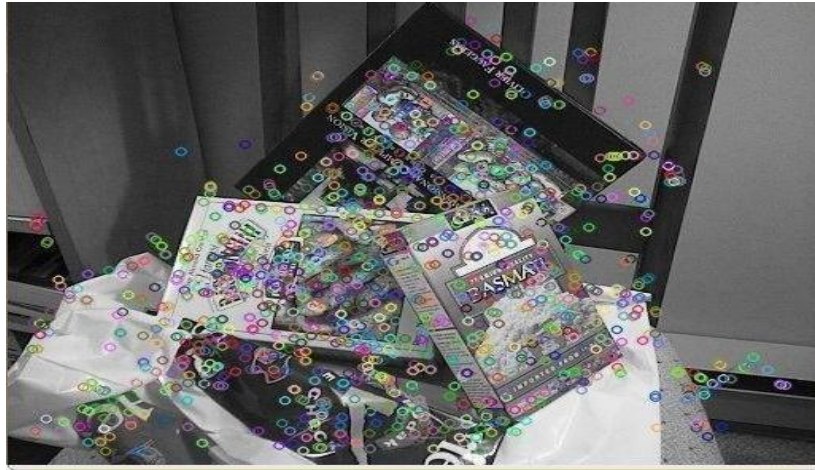


Figure 7: Example of Bag Of Visual Words

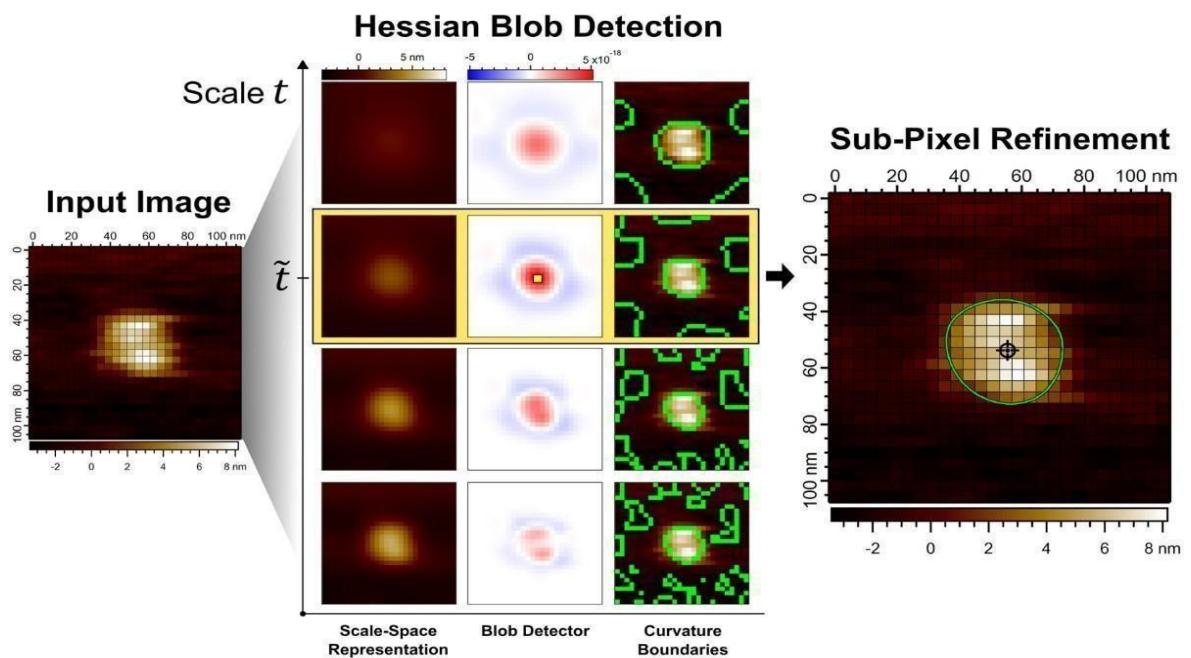


Figure 8: Hessian Blob Detection

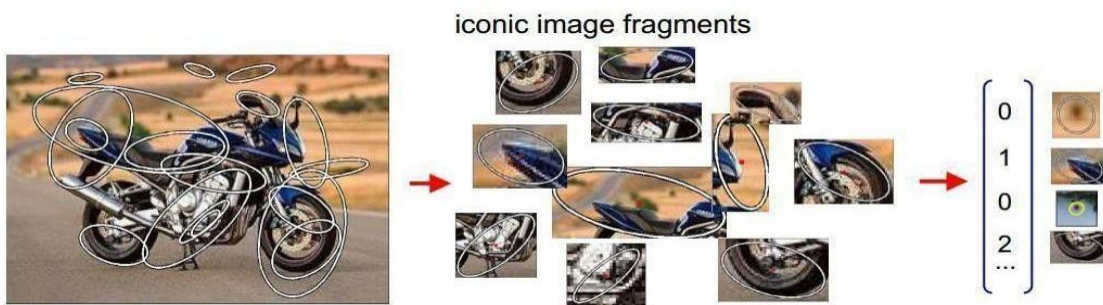
Feature-based traditional methods contributed to the advancement of computer vision systems. These methods found their applications in various fields such as object detection, face recognition, image retrieval, and landmark recognition. In Figure 8, blob detection algorithms like Hessian blob detection are used to detect blobs from images.

The interpretability of such methods was another key strength since the researchers could analyze the extracted features and the reasoning behind decisions made by the system. Another important aspect of these models was the low computational cost. The amount of computation needed for these methods is less than that required by deep learning models.

Nevertheless, feature-based traditional methods have multiple weaknesses. Firstly, the design of good feature descriptors is an arduous task. It involves significant research and experience. Moreover, even after investing substantial efforts, the feature extraction may not be generalizable for different domains or data sets. Secondly, traditional methods are incapable of capturing higher-level semantic meaning from the scene.

This problem of sensitivity to changes in the real world is yet another drawback. Lighting conditions, occlusions, angle of view, and other such factors can have an adverse impact on their performance. More often than not, the features obtained from the process are insufficiently resilient to withstand such changes.

Even more difficulties arise when working with satellite imagery. This is because of the complex scenes and a variety of objects, resolutions, and environmental noise in the images. The target landmarks may be only small fractions of the image size. Moreover, there is also the problem of atmospheric effects and sensor noise. In Figure 9, one may observe how the BoVW process works.



Converts an image into a vector, much like the bag-of-words model for text.

After that, you can apply standard ML techniques to build a classifier.

Figure 9: Bag of Visual Words Process

In this regard, classical feature-based approaches are frequently unable to ensure high efficiency and robustness when processing large-scale datasets. Thus, there was a necessity to develop more effective approaches allowing the automatic learning of features from data. Such approaches gave birth to deep learning solutions widely used nowadays in the field of image recognition.

Firstly, unlike feature-based approaches, deep learning algorithms enable to discover features at multiple levels, including low-level features like edges or texture as well as high-level ones like shapes and patterns. As a result, deep learning algorithms have been significantly improving the efficiency of image recognition systems by providing higher levels of performance [1].

Thus, it is possible to conclude that although feature-based solutions have greatly contributed to the development of image recognition technology and enabled significant progress in the field, there are certain limitations associated with them which allowed deep learning technologies to take the lead in computer vision. This shift is important not only from theoretical perspective but also plays a vital role in practical applications, e.g., monument detection from satellite images.

2.2 Deep Learning-Based Methods

As deep learning models become more complex, deeper, and capable, the question of model interpretability and transparency has taken center stage in the world of artificial intelligence. The latest neural network models, including CNNs, transformers, and hybrid models, have proven to be incredibly accurate in terms of output accuracy in a wide array of tasks [1]. The problem is that the process of making decisions can still remain opaque, and these "black box" models operate in such a way that makes understanding what goes inside hard or even impossible. This becomes a major issue in applications that require explanations for certain model actions.

In areas like cultural heritage protection, environment monitoring, urban planning, disaster management, and geospatial intelligence, prediction accuracy alone will never suffice. Researchers, policymakers, and experts in other fields will always want to know

precisely why an algorithm decided the way it did. For example, in recognizing historical buildings in aerial imagery, it is important to ensure that the decision is made according to architectural specifics and not based on anything else. Thus, the need for explainable artificial intelligence (or XAI for short) has become apparent in recent years.

As is evident from Figure 10, conventional approaches to feature extraction, like SIFT, allow for a structured workflow when analyzing image features.

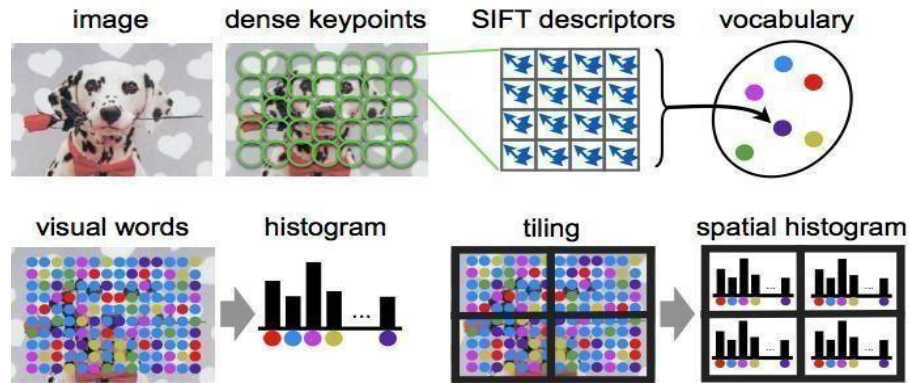


Figure 10: Working of SIFT

Explainable Artificial Intelligence is a field of study that aims at reconciling the efficiency of predictive models and human understanding. Specifically, it is concerned with designing methodologies and strategies that allow for gaining an insight into how a particular model works internally. In other words, XAI makes it possible to identify features or parts of an image that contribute the most towards the decision-making process. The ability to interpret and validate a prediction becomes vital in critical use cases when mistakes made by algorithms can have severe implications.

The need for explainability is even more pressing when working with satellite images. Firstly, it should be noted that they always include several overlapping elements. Namely, they may encompass vegetation, buildings, streets, water, shadows, and uneven terrains. Thus, the level of noise is relatively high, making it hard for a model to concentrate solely on the features that are relevant for the task. Consequently, the risk of identifying wrong structural patterns increases significantly without the means for validating predictions [10].

As can be seen from Figure 11, various techniques for class activation mapping have the ability to emphasize important areas which contribute to predictions.

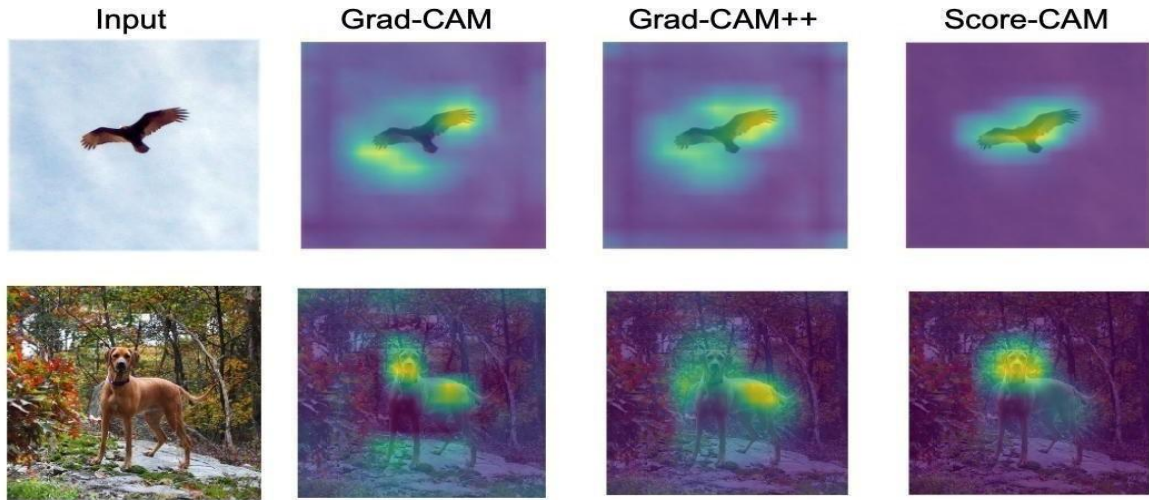


Figure 11: Grad-CAM , Grad-CAM, Score-CAM

Contextual bias is probably the most significant problem associated with remote sensing. The deep neural network learns associations between background objects and specific classes of monuments. For example, the deep neural network correlates green vegetation with some class of monuments only based on the training dataset containing samples with green vegetation around the target monument. While this may improve the accuracy on the training dataset, it can negatively affect the accuracy on unseen data. Therefore, it is essential to evaluate the behaviour of the deep neural network through the explainability methods.

It demonstrates a crucial weakness of data-driven models that they often learn spurious associations rather than real features of objects. This problem becomes relevant to remote sensing as data-driven models may misclassify monuments due to changing background, while the object stays the same. Using attention maps and saliency visualization, it is possible to verify whether the model learns features of the monument or irrelevant background.

Figure 12 shows an example of how Grad-CAM generates heatmaps for visualization of predictions.

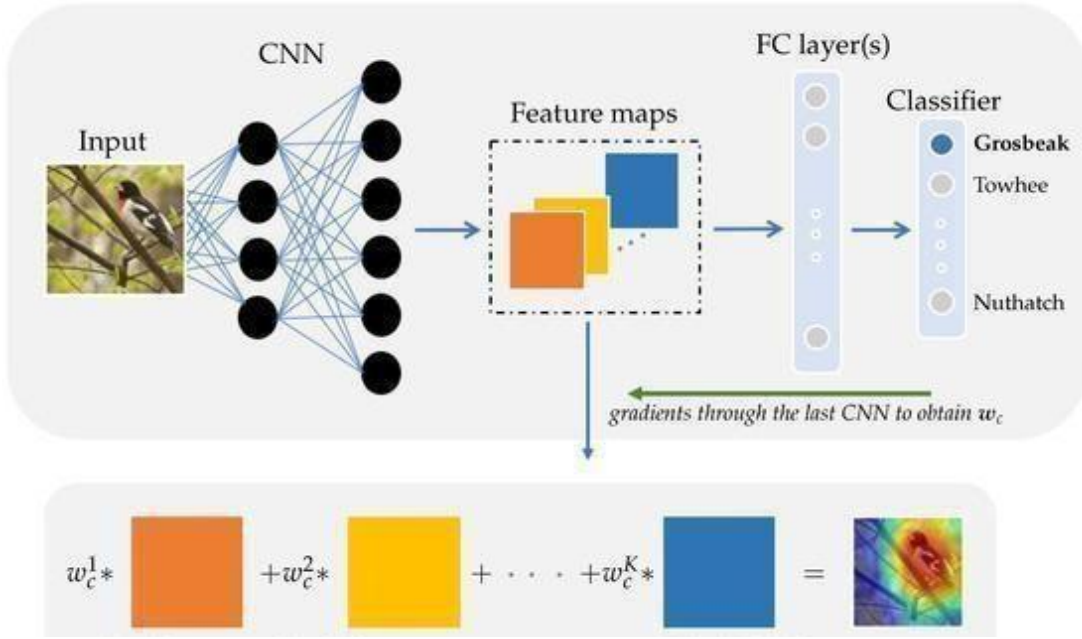


Figure 12: Gradient-weighted class activation mapping

The solution for this problem comes via explainable AI techniques that facilitate the visualization of the prediction. Since these techniques provide insight into the segments of the picture that have the greatest impact on the prediction, researchers can examine whether the model is learning from relevant segments. The relevance of the learning algorithm of the model is determined based on the comparison between the highlighted segments and the monument structure. If the model is learning from relevant segments, it is performing well, but in the case of irrelevant segments, the model exhibits bias.

The explainability of AI models helps researchers determine whether the decision-making process depends on relevant segments in the picture. By comparing these segments with the structure of the monument, one can understand whether the model is making predictions based on relevant information or irrelevant context. In case the highlighted segments include relevant information, the model can be considered reliable; otherwise, bias exists.

As illustrated in Figure 13, CNNs learn features hierarchically using convolutional layers and activation functions.

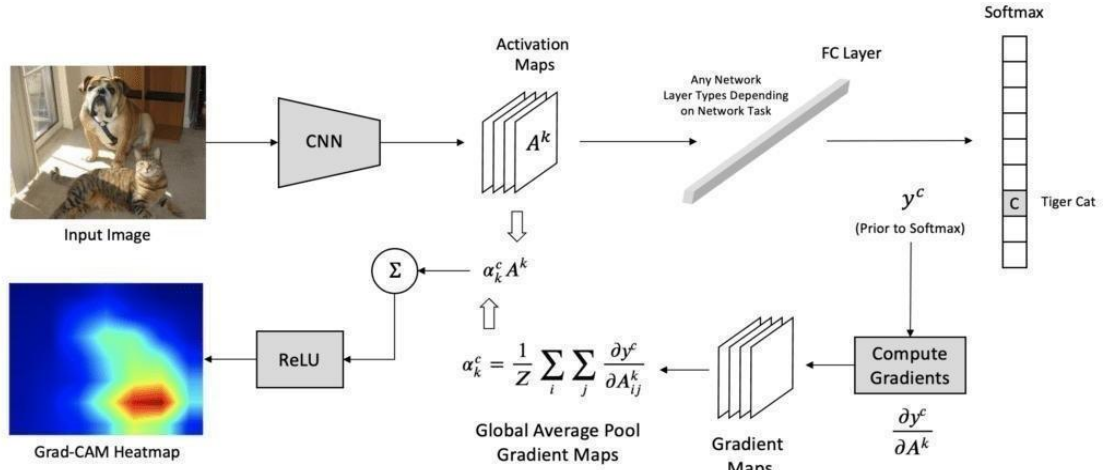


Figure 13: CNN Working& ReLU Working

Grad-CAM (Gradient-weighted Class Activation Mapping) [3] is one of the most common methods employed for explaining decisions made by deep networks. This method involves analyzing the gradients of the target class with respect to the feature maps of the final convolution layer in the CNN architecture. As a result, feature maps are combined into a weighted sum in order to produce a rough localization map. Grad-CAM is especially relevant to the domain of monument recognition since it allows us to make sure if the neural network depends on structural elements of the monuments in question, such as domes, arches, pillars, and layouts.

For instance, when Grad-CAM produces a heatmap highlighting the middle part of the dome on the picture, it means that the model uses relevant information in order to determine what kind of building it sees. If the localization map shows that the neural network focuses on greenery around the building, then it obviously uses some wrong information for classification purposes.

A popular approach for explaining predictions made by machine learning models is known as LIME (Local Interpretable Model-Agnostic Explanations) [4]. While Grad-CAM is a method used only with deep neural networks, LIME is not model-specific and can be used with any type of machine learning algorithm. LIME uses local modeling of the complicated machine learning model around a particular instance of input data. In LIME, input data, such as various regions of an image, are perturbed, and their influence on predictions is observed. From this, LIME extracts the areas most influential for the output.

Figure 14 shows the effect of LIME on predictions.

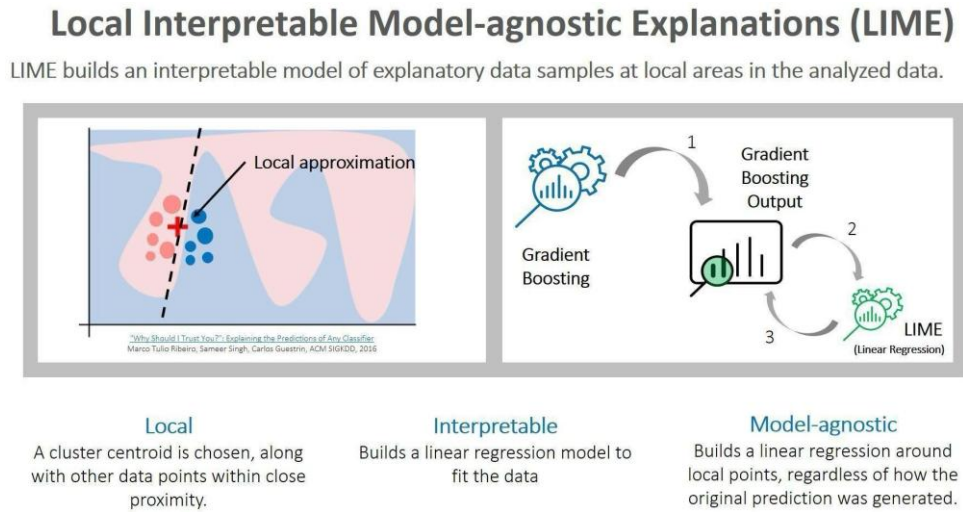


Figure 14: LIME

LIME is highly effective when it comes to explaining predictions at an individual level. For example, in the case of multiple objects present in the satellite imagery, LIME will help point out the particular parts of the picture that affected the model's decision regarding the monument in question. This kind of explanation offers useful insights about the model's thinking process and may help spot any possible mistakes or bias.

One more technique worth mentioning is SHAP (Shapley Additive explanations) [5]. SHAP draws upon game theory principles and computes the importance of every single feature according to its contribution to the model's output, assigning Shapley values. The advantage of SHAP is that it offers not only local but also global explainability, allowing users to study predictions individually and evaluate model performance as a whole.

Among other benefits, SHAP has a sound theoretical basis that guarantees fair and consistent attributions of features' contributions to the model's prediction. This means that SHAP may be applied to determine the role of particular pixels in the image analysis and classification task.

The incorporation of XAI methods with deep learning models has gained increased significance in remote sensing applications. With the help of high-performing models and XAI methods, scientists can build models that are not only reliable but also highly accurate. XAI helps validate the behavior of the models to ensure that predictions are

made on important features rather than spurious correlations [12].

Moreover, XAI techniques help improve the performance of models. Through analysis of XAI-based explanations, scientists can understand where the problems lie within their models. If the model keeps focusing on the background parts instead of the relevant part of an image, it might be necessary to implement advanced image preprocessing steps such as saliency-based cropping and better data augmentation techniques.

Figure 15 shows the importance of different features for model predictions using SHAP.

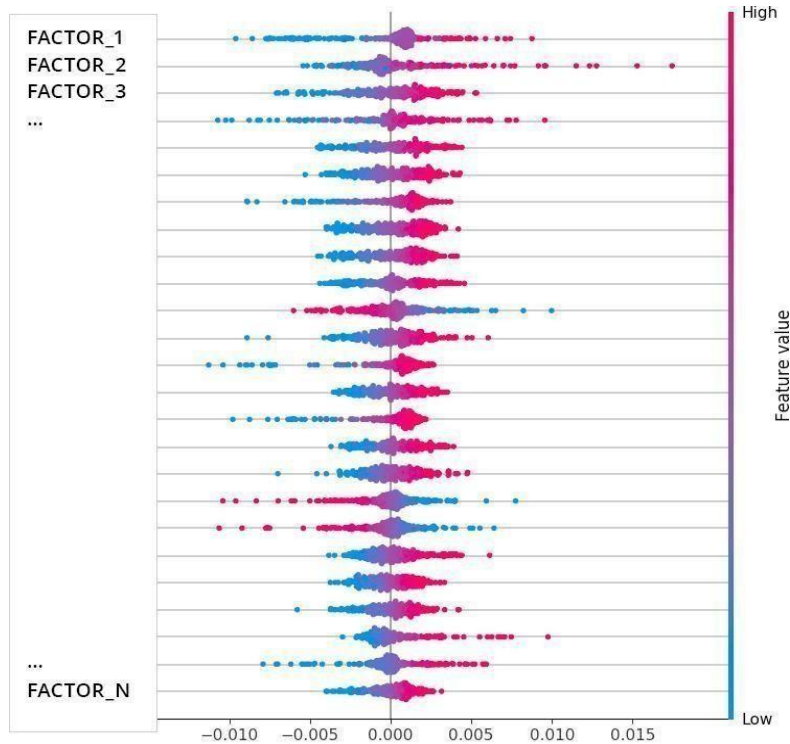


Figure 15:SHAP Value

Although there are many benefits associated with explainable AI, there are also some challenges that need to be considered. The creation of explanations may be a resource-intensive process, especially when applied to large and complicated models. Moreover, various approaches to XAI may yield different explanations for the same prediction. Finally, there is a need for further research into evaluating the quality of explanations.

To conclude, Explainable Artificial Intelligence can be regarded as one of the most significant milestones in the history of artificial intelligence. Unlike classical black-box

systems, Explainable Artificial Intelligence is capable of providing users with valuable insights into how predictions are made. It is possible to ensure that decisions are based on relevant and important variables. In the context of remote sensing and monument recognition, the use of various explainable AI tools such as Grad-CAM [3], LIME [4], and SHAP [5] increases the credibility and acceptance of AI applications.

2.3 Summary of Literature

With advancements in deep learning techniques that have continued to increase in terms of complexity and efficiency, the need for interpretability and transparency has become crucial in artificial intelligence. The use of architectures like Convolutional Neural Networks (CNNs), transformer models, and even hybrid models enables high accuracy on many problems [1]. However, the models work as “black boxes,” meaning that while they perform exceptionally well, the logic used to achieve this kind of performance cannot be comprehended easily. There are numerous challenges brought about by the inability of such models to provide any form of explanation. In some cases, there is a need for transparency and accountability in decisions made using such models. In scenarios like cultural preservation, environmental surveillance, city planning, disaster prevention, and intelligence, it is important to go beyond prediction accuracy and understand why an architecture makes certain decisions. This means that, for example, when trying to identify a monument using satellite images, it must be made sure that it is the architectural structure and not something else in the image that determines the classification. The increasing demand for interpretability has resulted in the development and swift progress in Explainable Artificial Intelligence (XAI).

Explainable Artificial Intelligence seeks to overcome the problem of reconciling advanced machine learning models with human comprehension. The concept centers around the development of methodologies and technologies that shed light on the inner workings of the models. By highlighting the factors or areas of the input data responsible for driving a certain prediction, XAI allows for the interpretation, validation, and confidence in the outputs generated by the models. This aspect is crucial when considering the potential repercussions of inaccurate or biased predictions.

When dealing with satellite imagery, the need for interpretability becomes even more pressing owing to the complexity of the data. Satellite images tend to have multiple overlapping elements such as vegetation, constructions, road networks, water surfaces, shadows, and terrain changes. Such overlapping makes it rather hard for the algorithm to concentrate on the essential aspects of the image. Without interpretability, it is impossible to evaluate whether the model's decisions were influenced by essential structure-based factors or random artifacts of the background environment [10].

Contextual bias in the form of correlation between other elements present in the background and classes is another problem that can affect remote sensing. The deep learning model can learn that a certain monument is always surrounded by dense vegetation, for example, and use this correlation to make decisions in favour of one class or another. This approach can yield positive results during training but fail in situations when such correlation does not exist anymore. The issue mentioned above shows the necessity to verify the model behaviour with explainable techniques.

Explaining the decisions made by artificial intelligence becomes possible through the utilization of various approaches known collectively as Explainable Artificial Intelligence (XAI). Using these tools allows to detect which portions of an image were crucial for making a decision about the structure in question. In this regard, if the selected portions are indeed related to the monument's actual structure, it can be assumed that the deep learning model works effectively. Otherwise, it means that there is some kind of bias or error.

One of the most commonly adopted explainability methods in deep learning models is Grad-CAM (short for Gradient-weighted Class Activation Mapping) [3]. Grad-CAM is based on the principle of examining the gradients corresponding to the target class concerning the final convolutional layer's feature maps. Based on these gradients, a weighted sum of the feature maps can be constructed, yielding a localization map that shows areas of importance. The method outputs a heatmap applied to the input image.

This tool becomes especially valuable when dealing with monument recognition problems, allowing us to check whether the model attends to any structural elements of

the monument, such as domes, arches, pillars, or layout. For instance, if the heatmap emphasizes the center of the dome of the monument, we can conclude that the model uses appropriate visual features in its decision-making process. Otherwise, it means that the model attends to irrelevant cues, like surrounding plants or roads.

Another important method for explainability is the Local Interpretable Model-Agnostic Explanations (LIME) [4]. While Grad-CAM is only applicable to deep learning algorithms, LIME is not bound to any algorithm but can be used on any machine learning model. LIME approximates the complex model locally around the given input point. It introduces various changes to the input point, such as obscuring some areas of the image, and analyses how the output of the model was affected. Based on the result, LIME determines which areas contributed most to the model's prediction.

LIME is especially valuable when trying to understand the reasoning behind a single prediction. For example, when using a satellite image containing several objects, LIME will show which parts of the image contributed to the prediction of a certain monument.

A third approach with great potential and strong theoretical underpinnings is SHAP (Shapley Additive explanations) [5]. As suggested by the name, SHAP is rooted in cooperative game theory and employs Shapley values, i.e., scores of feature importance in respect to their impact on the value produced by the model. As mentioned before, SHAP supports both global and local analyses of the predictions made by the machine learning algorithm.

The strongest advantage of the SHAP approach is its rigorous theoretical foundation, implying fairness and consistency of feature importance attribution in the model. In image analysis, SHAP can be used to assess the impact of the region or even pixel in question on the classification process. The combination of deep learning and XAI approaches opens new opportunities for remote sensing applications.

As previously stated, the combination of high-performance deep learning models and XAI approaches allows achieving not only excellent prediction accuracy but also providing a way to validate model behavior. This implies that predictions are made based on the correct features in the dataset [12], rather than random spurious correlations that

can appear due to chance. This capability is especially critical for monument recognition as it is necessary to make sure predictions are correct.

In addition to that, another important use of XAI approaches is in designing better machine learning models and improving their performance. Using the explanations obtained through XAI techniques, scientists can determine the areas of weakness, biases, or faults within the model and find ways to overcome them. For instance, the tendency of a certain model to focus too much on unimportant background parts would mean that improvements should be made in regard to image cropping or more efficient data augmentation strategies.

Even though it offers plenty of benefits, explainable AI is faced with a number of issues. To begin with, the process of generating explanations is highly computationally costly, especially when applied to more complicated models. Furthermore, different methods of generating explanations might provide inconsistent results. Thus, evaluation of the generated explanations is one of the problems that still need to be addressed since no common criteria have been developed yet.

Explainable Artificial Intelligence can be regarded as an important breakthrough in the field of artificial intelligence as this technology helps solve all the shortcomings associated with traditional black-box models through transparency, accountability, and explainability, ensuring that the predictions made are based on appropriate features.

To conclude, Explanation for Artificial Intelligence is a breakthrough in the field of artificial intelligence because this technique improves on all the limitations that come along with traditional black-box models through transparency, interpretability, and accountability. The technique helps make sure that model predictions are based on meaningful features.

CHAPTER 3

PROPOSED METHODOLOGY

The methodology proposed within the scope of this project is based on the design and implementation of an intelligent framework that will help us recognize monuments using satellite images. This intelligent framework is named IndiSight++ and utilizes the latest achievements in deep learning along with Explainable Artificial Intelligence (XAI), allowing us to build accurate models that are interpretable as well [1], [3]–[5]. While the vast majority of the existing approaches focus only on classification quality, we propose to make sure that our predictions are also comprehensible.

IndiSight++ represents a multi-stage pipeline comprising data preprocessing, features' extraction via deep learning, classification itself, and post-hoc analysis based on XAI techniques. All stages were chosen considering the peculiarities of satellite images' analysis, including the problem of background noise, different sizes and scale, as well as insufficient labeling [10]. Features' extraction stage is realized through the application of a lightweight convolutional neural network that allows efficient satellite image analysis due to high computational performance [2]. Finally, the application of post-hoc explanation methods (Grad-CAM, LIME, and SHAP) makes our approach interpretable as well [3]–[5].

3.1 Model Selection: MobileNetV1

Selection of appropriate architecture plays an essential role in the effectiveness and efficiency of the system. The main deep learning algorithm employed in this project is MobileNetV1, owing to its light-weight nature [2].

Conventionally, convolution neural networks are made up of millions of parameters that are computationally expensive and, therefore, cannot be applied in satellite image processing. To overcome this challenge, MobileNetV1 uses a depth-wise separable convolution mechanism that reduces computation costs while still providing feature extraction capabilities.

A standard convolution can be decomposed into two processes as follows:

Depth Wise Convolution – This process applies filters on each channel of the input data separately to capture spatial information.

Point Wise Convolution (1×1) – The outputs from the previous process are combined together to generate feature maps.

To gain insights into the computational efficiency of this method, the formula for a standard convolution can be defined as follows:

$$C_s = D_k \times D_k \times M \times N \times D_x \times D_x$$

where:

D_k = Kernel size

M = Number of input channels

N = Number of output channels

D_x = Spatial size of feature maps

On the other hand, Depth-wise Separable Convolution breaks this down into two stages:

Depth-wise Convolution:

$$C_d = D_k \times D_k \times M \times D_x \times D_x$$

Pointwise Convolution:

$$C_p = M \times N \times D_x \times D_x$$

Total computational complexity:

$$C_{total} = C_d + C_p$$

With this type of decomposition, the number of parameters as well as floating point operations (FLOPs) is reduced, making MobileNetV1 extremely efficient. This efficiency can be estimated by the following expression:

$$\text{Reduction} \approx (1 / N) + (1 / D_k^2)$$

It can thus be observed that MobileNetV1 greatly reduces computational complexity in relation to standard Convolution Neural Network, without compromising accuracy [2]. In other words, it is particularly suitable for real-time processing and for tasks such as analysis of satellite images.

In addition to that, MobileNetV1 also provides options of hyperparameters such as width and resolution multipliers.

As can be seen from Figure 16, MobileNet has replaced conventional convolutional layers with depth-wise and point-wise convolutions, allowing for efficient feature extraction at lower computational cost.

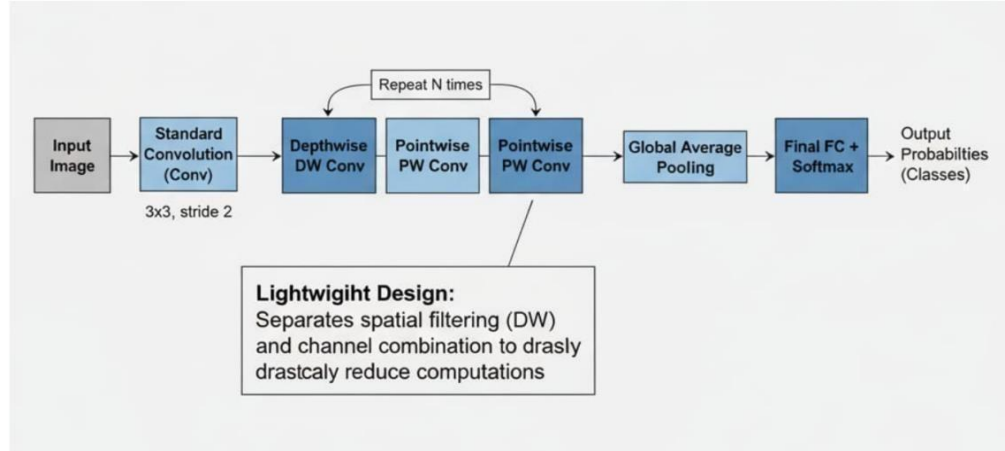


Figure 16: MobileNet architecture

3.2 System Architecture

The overall architectural design of the system has been carried out in a systematic manner in accordance with the pipeline design. The architecture of the suggested IndiSight++ model comprises several consecutive stages such as acquisition, preprocessing, deep learning-based feature extraction, classification, and XAI-based post-hoc explainability. Each stage has been carefully thought of to deal with issues such as background clutter, scale differences, and complex visual textures that may appear in satellite images [10].

3.2.1 Input Layer

Firstly, satellite images act as the main data source for the proposed system. These data sources can be obtained from publicly available geospatial databases and can have a high degree of variability when it comes to factors like resolution, scale, lighting, and environmental conditions. Hence, proper preprocessing is necessary before introducing these images as input data. Satellite images provide the capability to monitor and analyze monuments that are geographically distributed, making satellite images an ideal data source for remote sensing [10].

3.2.2 Initial Convolution Layer

Convolution is the operation used by the first layer of the neural network for extracting low-level features such as edges, gradient, and textures. Low-level features form the basis for all complex higher-level features to be extracted by the neural network, and thus by obtaining these fundamental features, the first layer helps to facilitate learning by the following layers [1].

3.2.3 Depth Wise Separable Convolution Blocks

The backbone architecture is based on several depth-wise separable convolution layers, which are the basic building blocks of MobileNetV1 [2]. By separating standard convolution into depth-wise and point-wise convolutions, the blocks can efficiently extract features while decreasing the complexity of computations.

Convolution layers hierarchically learn abstract feature representations at different abstraction levels:

- Basic levels learn simple features, such as edges and texture
- Intermediate levels learn shapes and structural features
- Advanced levels learn semantic information, such as monuments' configurations and architectural designs

This ability to learn multi-level features helps the network understand complex visual

structures contained in satellite images, ensuring accurate classification even when environmental factors affect the image's background.

3.2.4 Global Average Pooling Layer

Fully connected layers require numerous parameters for computation. Instead, global average pooling (GAP) is performed to decrease the size of the feature maps. The method calculates the mean value of each feature map, converting them into one value per channel [1], which greatly reduces the number of parameters and prevents overfitting.

There are numerous benefits of using the proposed method:

- It prevents overfitting by removing large fully connected layers
- It is computationally efficient and saves memory space
- It keeps spatial data intact since it maintains the correlation between feature maps

The output layer employs the softmax activation function to generate probability values corresponding to the monument categories. Softmax normalizes the output logits to obtain a probability distribution whose total value sums to one. The category with the highest probability score becomes the predicted output.

3.3 Transfer Learning

For better results and shorter training duration, transfer learning is used. Pre-trained weights from the MobileNetV1 architecture are applied, which have been trained on the ImageNet dataset consisting of millions of images belonging to thousands of classes [2].

Pre-trained weights help the model utilize prior knowledge of features such as edges, textures, and shapes, as they are transferable in nature and can be easily applied in any domain.

The transfer learning algorithm includes:

- Fixing earlier layers to preserve general features

- Fine-tuning deeper layers to capture monument-specific features
- Adding an appropriate classification layer according to the number of monument classes

The application of this technique greatly alleviates the problem of the need for large labeled data sets and speeds up the convergence speed during the training procedure. Moreover, transfer learning makes a better generalization performance, especially with domain-specific data limited to the available amount.

3.4 Data Preprocessing and Augmentation

Preprocessing of the data is an important stage that allows to improve generalization and model quality. The following steps are used:

3.4.1 Image Resizing

Each image from the dataset is resized to the specific size suitable for MobileNetV1 (e.g., 224×224 px). It allows having the same image size for further training and applying batching techniques for processing. Also, resizing is useful for keeping consistent feature extraction among images.

3.4.2 Normalization

Pixel values of each input are rescaled to the specific interval (e.g., [0,1] or [-1,1]), thus providing for faster convergence during training [1].

3.4.3 Data Augmentation

Some of the methods used to increase diversity and avoid overfitting include:

- Rotation (for handling different orientations)
- Flipping horizontally and vertically
- Scaling and zooming
- Random cropping

- Adjustments of brightness and contrast

These augmentation methods provide a means for simulating real-life differences within satellite pictures, which help the model generalize and improve its robustness to new images.

3.4.4. Saliency-Based Cropping

The use of a saliency-based cropping technique has been introduced as one of the novel techniques in this methodology aimed at mitigating the issue of background prominence. In general satellite imagery, a significant portion of the picture consists of information that is irrelevant to the task at hand, including vegetation, roads, water bodies, and other urban infrastructure while the monument might be present as a tiny, unremarkable area.

This challenge can be addressed using saliency techniques that can detect areas of the picture that are more relevant to the prediction process [8], [9]. The saliency maps contain attention maps generated based on intensity variations and structural relevance.

Once the salient areas have been detected, dynamic cropping of the image takes place, resulting in the exclusion of all irrelevant backgrounds, thereby reducing unnecessary information and ensuring that only the important parts remain. Thus, the method ensures improved classification accuracy and minimizes the influence of context bias while analyzing satellite imagery.

3.5 Explainable AI Integration

The main innovation introduced by the presented methodology is the integration of Explainable Artificial Intelligence (XAI) techniques for interpreting and verifying the predictions made by the model. Although deep learning algorithms provide accurate results, the decision-making process within them remains opaque. To overcome this problem, several approaches to XAI are incorporated in the presented framework. These techniques ensure that the model's predictions are based on relevant features extracted from satellite imagery [3]-[5].

The following XAI techniques are implemented in the project:

3.5.1 Grad-CAM

Grad-CAM is an algorithm generating heat maps showing which parts of the image contribute to the prediction made by the model [3]. Grad-CAM works on the basis of computing gradients in the final convolution layer of the neural network and generating a localization map of the most significant features. It verifies that the model bases its decision on meaningful architectural structures of the monuments, not on irrelevant background features. As can be seen in Fig.17, heatmaps produced by Grad-CAM offer a clear understanding of the model's attention mechanism.

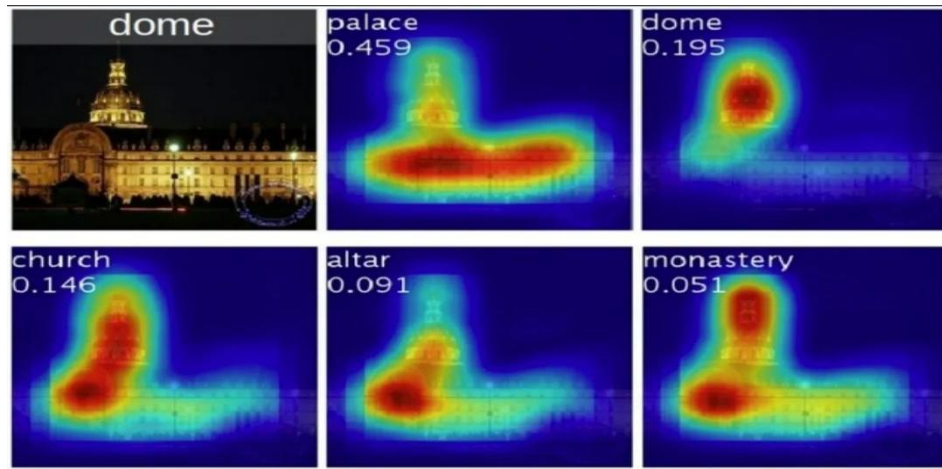


Figure 17: Grad-CAM Heatmap

3.5.2 LIME

LIME works by creating a dataset of perturbed samples around the chosen data point and measuring the change in the prediction of the model when the samples are fed into the model. In image data, this is normally achieved by segmenting the image into smaller units known as superpixels and toggling some of the superpixels. The model's prediction is calculated for each perturbed sample and then an interpretable model is built using the perturbed samples and weighted based on the distance from the original sample.

The explanations obtained from the LIME algorithm indicate the regions that positively or negatively impact the prediction. Therefore, we are able to determine whether the model is using relevant information in making predictions or irrelevant information like the background noise. Furthermore, LIME is model agnostic and thus can be used with

any model irrespective of its architecture and parameters.

3.5.3 SHAP

Explanation using SHAP utilizes cooperative game theory where each feature is assigned a score indicating its importance [5]. The SHAP method uses Shapley values to find out the contribution of each individual feature towards the final prediction. The SHAP explanation framework supports both local and global explanations, enabling predictions and model behavior analysis.

Through these methods, it becomes possible to ensure that the decisions made by the model are transparent, reliable, and depend on valid visual cues.

3.6 Training Procedure

The training process consists of several stages for successful model learning and generalization:

- The dataset is split into train, validate, and test sets for the appropriate evaluation of the model.
- The model is trained by minimizing the categorical cross-entropy loss function that quantifies the discrepancy between the predicted and true class labels.
- Efficient weight updates can be done with optimization techniques such as Adam and Stochastic Gradient Descent (SGD).
- Model quality is assessed using accuracy, precision, recall, and F1 score metrics.
- Methods like dropout and early stopping are employed to avoid overfitting and ensure generalization [1].

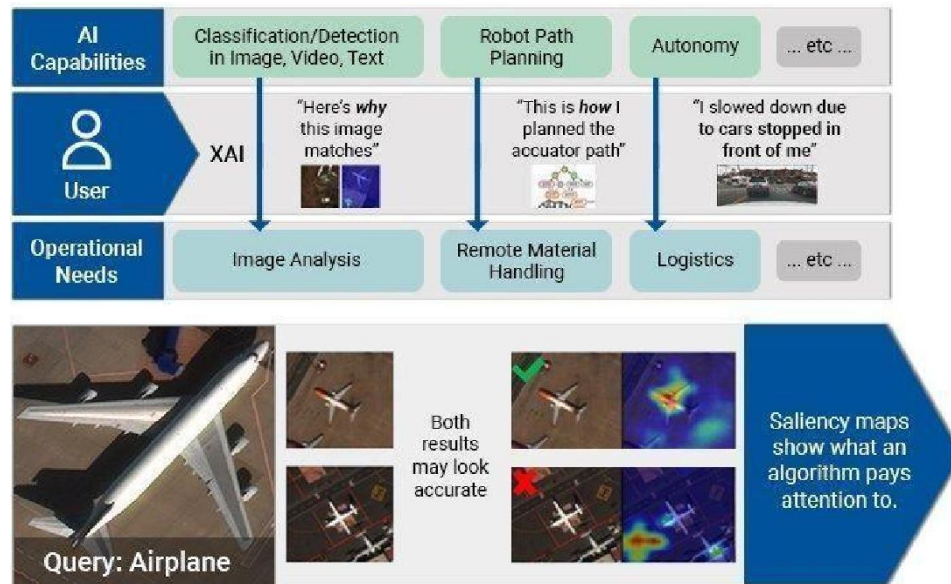


Figure 18: Overview of Explainable AI

Figure 18 clearly illustrates that Explainable AI facilitates model interpretations.

3.7 Advantages of the Proposed Approach

Some significant advantages of the proposed approach are listed below:

- **Computational Efficiency:** Due to the lightweight MobileNetV1, there will be fewer computational resources consumed [2].
- **Accuracy:** Deep Learning makes feature extraction more effective and hierarchical [1].
- **Generalization:** Through data augmentation, better results can be achieved on unknown data.
- **Prevention of Overfitting:** Through transfer learning and preprocessing, stability can be improved [10].
- **Explainability:** Explainable AI techniques ensure the interpretability of models' decisions [3]-[5].
- **Scalability:** The proposed approach can be used for analyzing massive satellite images [10].

- **Reliability:** Decisions made by the models will be more reliable through explanations.

3.8 Summary of Methodology

To conclude, the proposed IndiSight++ methodology is an optimal mixture of effectiveness, accuracy, and interpretability. Specifically, the employment of the MobileNetV1 algorithm guarantees low computing costs, whereas transfer learning significantly improves the performance even when the data volume is restricted. In addition, state-of-the-art preprocessing approaches, such as saliency-based cropping, are utilized to optimize feature extraction and remove background interference. Finally, the inclusion of XAI techniques, namely Grad-CAM, LIME, and SHAP, guarantees model interpretability and explainability [3]–[5].

Therefore, the developed methodology can be successfully applied in practical applications involving the detection of monuments from satellite imagery.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Experimental Results and Discussion

In this chapter, there will be an elaborate evaluation of the proposed system with regard to the ability to classify monuments, training procedures, and decision making by using explainable artificial intelligence (XAI) techniques. The aim is not only to analyze how accurate this model works, but also the stability, adaptability, and transparency.

The experiment will be conducted on an appropriately prepared set of satellite pictures representing several categories of monuments. The assessment methodology involves quantitative evaluation by measuring accuracy, precision, recall, and the F1 score and qualitative assessment by means of visual explainability techniques.

4.2 Training Strategy and Model Design

The model for the proposed system is designed based on the concept of transfer learning and uses MobileNetV1 as the base model [2]. This model allows the use of knowledge gained during the training process on the large-scale dataset of ImageNet.

ImageNet is an extremely large database with millions of pictures classified into hundreds of different categories, and therefore this dataset enables the learning of generic image features like edges, texture, gradients, and shape. Initializing the model with pre-trained weights ensures that the system achieves faster convergence and better results.

In order to make the network suitable for monument classification, changes were made to the network architecture through replacement of the final dense layer with a classification layer specific to the number of monument categories.

The training of the model had the following parameters:

- **Optimizer:** Adam optimizer
- **Loss function:** Categorical cross entropy
- **Batch size:** Based on hardware constraints
- **Learning rate:** Optimized for stable convergence

For validation of the results and to avoid overfitting, the dataset was split into:

- **Training dataset:** for learning model parameters
- **Validation dataset:** for hyperparameter tuning
- **Test dataset:** for final evaluation on unseen data

4.3 Model Performance and Accuracy

The designed system had an overall classification accuracy of around 92.7%, which proves the efficiency of the deep learning approach.

The Accuracy can be calculated as follows:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Where TP, TN, FP, and FN denote true positive, true negative, false positive, and false negative, respectively.

Other performance indicators, aside from accuracy, were considered:

- $\text{Precision} = TP / (TP + FP)$
- $\text{Recall} = TP / (TP + FN)$
- $\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

The model demonstrated impressive results across all these performance indicators, ensuring balanced performance without any preference for certain classes.

The application of transfer learning greatly helped improve performance through faster convergence and enhanced feature extraction [2].

4.4 Training and Validation Analysis

Training and validation curves reveal a great deal about model learning patterns.

Figure 19 shows that training and validation accuracy grow steadily as the number of epochs increases, which suggests efficient feature learning.

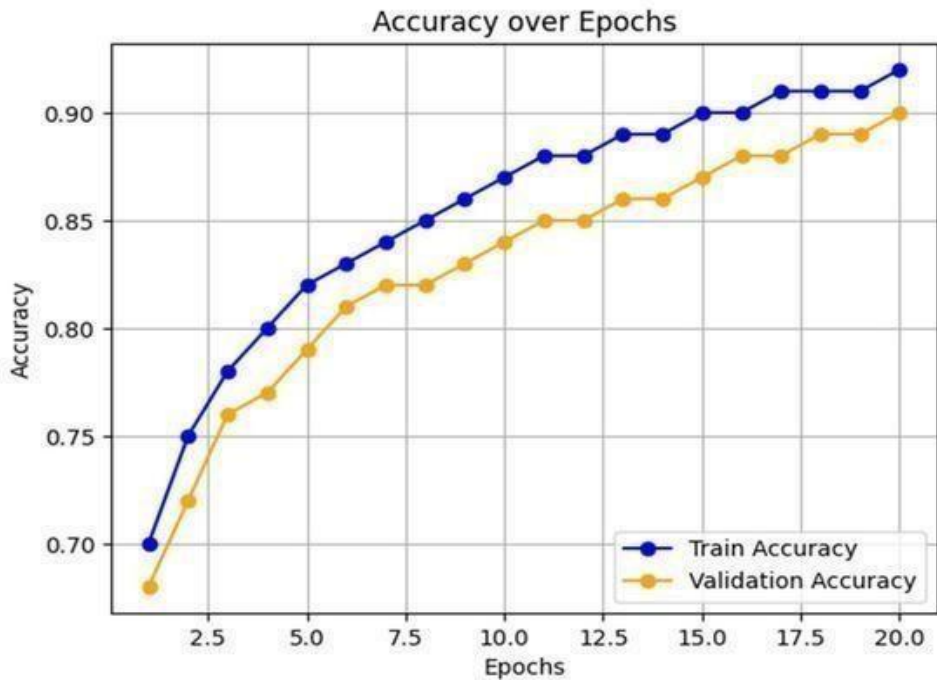


Figure 16: Training vs Validation accuracy over epochs

From Figure 20, it is evident that there is consistent reduction in the losses of training and validation, suggesting consistency in optimization.



Figure 17: Training & Validation loss

There is a minimum difference between training and validation accuracy, which suggests that there is no problem of overfitting in the model.

Figure 21 demonstrates underfitting and overfitting.

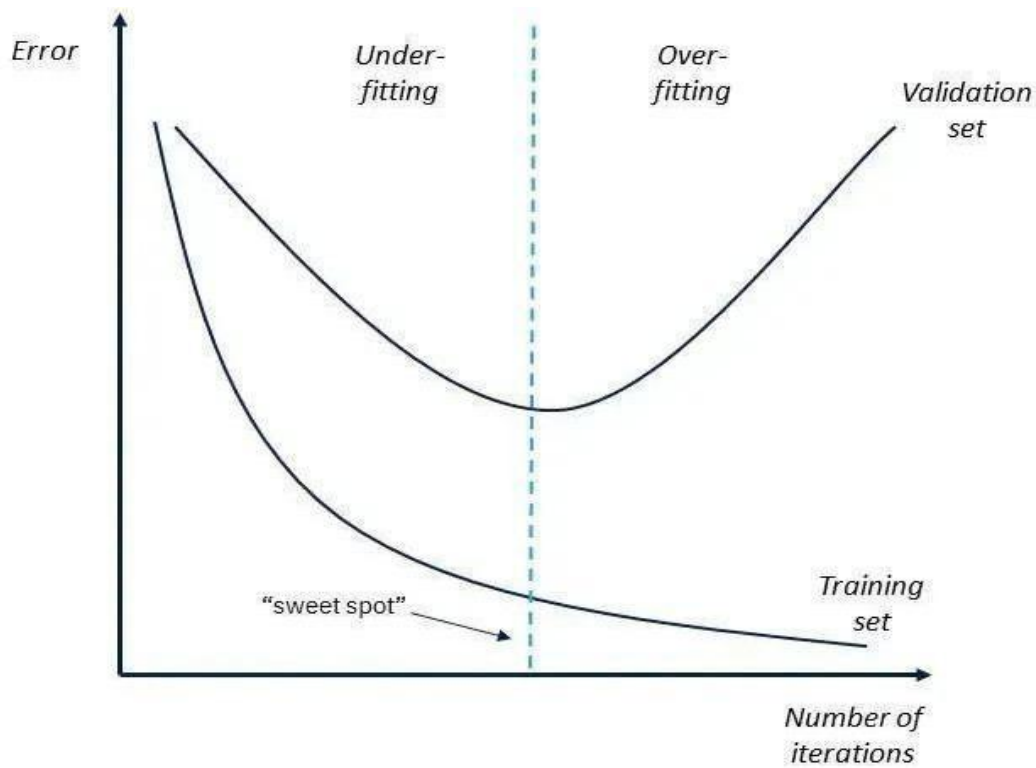


Figure 18: Illustration of Underfitting & Overfitting over Training & Validation set

The following approaches have proven effective in tackling:

- **Overfitting** – managed by using dropout and augmentation
- **Underfitting** – prevented by ensuring adequate model capacity
- **Training instability** – stabilized using adaptive optimization

4.5 Explainability Results

An important aspect of this paper is the application of XAI methods [3]–[5].

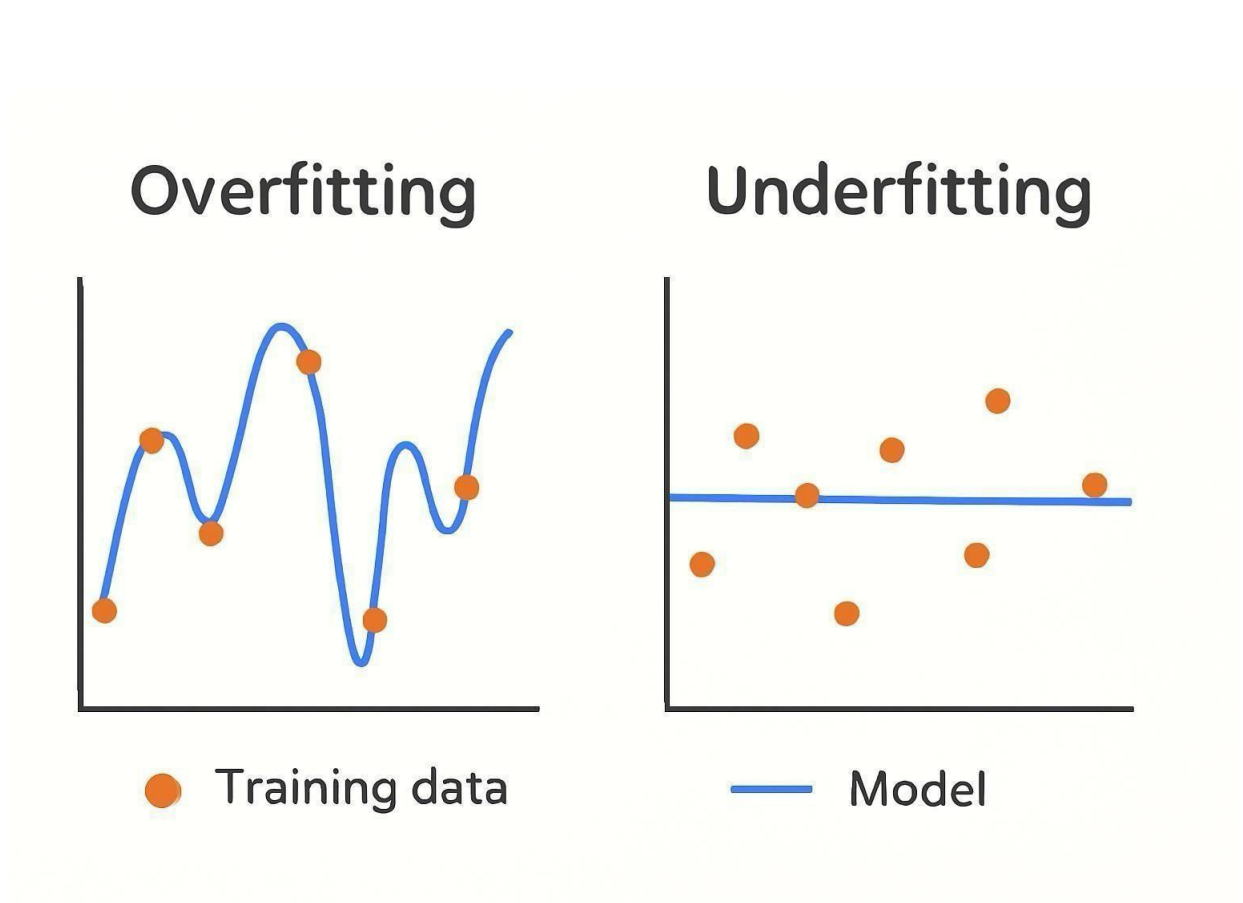


Figure 19: Comparison of Overfitting & Underfitting

4.5.1. Grad-CAM Analysis

Grad-CAM creates heat maps of significant areas affecting the predictions [3]. The results prove that the model concentrates on structures, including domes, arches, and spatial arrangements.

4.5.2. LIME Analysis

LIME produces a local explanation of each prediction [4]. This proves that the predictions are made using relevant sections of monuments.

4.5.3. SHAP Analysis

SHAP estimates the impact of features on predictions [5]. It ensures local and global consistency in interpretations.

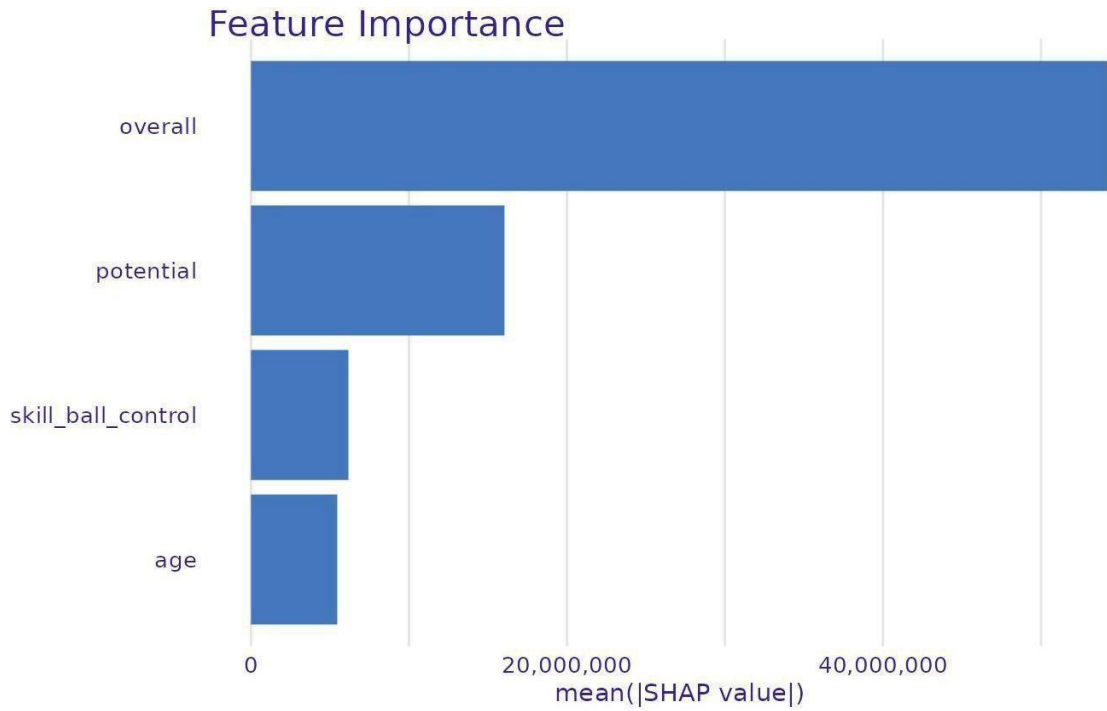


Figure 20: Feature Importance using Shap values

4.6 Discussion of Results

The experimental findings show that the proposed framework successfully balances efficiency, accuracy, and interpretability.

4.6.1 Advantages of the Proposed Framework

- High level of classification accuracy (~92.7%)
- Efficient computations based on MobileNetV1

- Excellent generalization performance
- Transparency provided by XAI

Due to the lightweight structure of the framework, it can be applied to real-world applications.

Due to the lightweight design of the proposed model, it can be easily implemented in real-world scenarios, even with limited computing power.

The introduction of XAI techniques helps solve one of the main drawbacks of deep learning – opacity. The provision of visual interpretations allows the model to become more reliable and user-friendly.

4.6.2 Disadvantages

However, there are several disadvantages to the proposed system:

- Sensitivity to the quality of the input data
- Reduced accuracy when detecting small and occluded monuments
- Resolution issues in satellite images
- Extra computational overhead due to XAI.

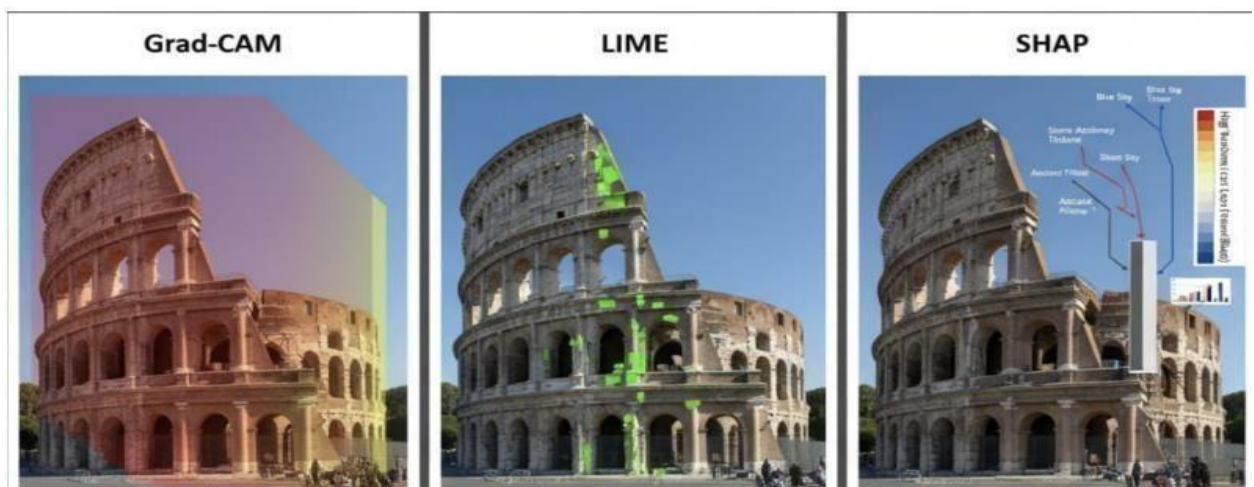


Figure 21: Grad-CAM vs LIME vs SHAP

4.7 Summary of Results

Experimental verification of the proposed architecture validates its efficacy. This model exhibits:

- Highly accurate and balanced performance
- Consistent training and validation patterns
- Excellent generalization capabilities
- Reliable and interpretable predictions

The combination of deep learning and explainable artificial intelligence constitutes a robust approach for monument detection based on satellite images.

Some possible use cases include:

- Heritage conservation
- Remote sensing
- Geographic information systems

Further research directions could involve expanding the dataset variety, increasing model robustness, and developing advanced explainability methods.

4.8 Comparative Analysis and Model Robustness

In order to verify the effectiveness of the proposed IndiSight++ architecture, it is necessary to conduct a comparative analysis that highlights its robustness under different conditions. Satellite images pose many challenges related to resolution, illumination, seasonal changes, and complex backgrounds. Nevertheless, the proposed method performs consistently well. The reason lies in the application of data augmentation and transfer learning. The former enables the model to extract invariant features, whereas the latter facilitates faster convergence and better generalization on new data.

Another important component of the system is its capability to concentrate on the salient

parts of the input image and ignore irrelevant background details. Typically, satellite pictures comprise several different components, including vegetation, roads, and the infrastructure of the city itself. By integrating saliency-based cropping, the potential bias in predictions due to irrelevant information present in the image can be reduced. In other words, using this approach allows the system to detect monument-specific features and ignore all others in order to increase the classification accuracy and robustness.

Finally, another crucial factor in analyzing the performance and robustness of our proposed solution is its interpretability. The implementation of the Explainable AI methods such as Grad-CAM, LIME, and SHAP makes it possible to investigate how the neural network processes input data and why it makes a particular decision. As can be seen from the visual representations, the algorithm pays attention to architectural features such as domes, pillars, and layouts.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This work discussed the development of IndiSight++, a novel monument recognition framework which is intended to automatically identify and classify Indian monuments based on satellite images. It can be considered as a groundbreaking contribution towards the integration of modern deep learning approaches and Explainable Artificial Intelligence (XAI) in order to provide both efficient predictions and interpretability. At the same time, this approach tackles a crucial shortcoming of existing solutions, namely the lack of interpretability, which hinders effective model validation and decision-making processes [3]–[5].

One of the primary advantages of the proposed solution is that it combines the benefits of accuracy, interpretability, and efficiency into a single framework. Due to the use of the lightweight model based on MobileNetV1, the system manages to attain efficient feature extraction, while keeping computational complexity relatively low [2]. Contrary to the traditional convolutional neural networks that demand considerable computational efforts and resources, MobileNetV1 uses depthwise separable convolutions which allow significantly lowering the number of parameters and operations required. Therefore, the proposed model proves to be highly efficient in processing large satellite images.

The inclusion of transfer learning improves the model's efficiency. Initialization of the model using pre-trained weights derived from massive datasets like ImageNet helps to leverage existing knowledge regarding common visual features such as edges, textures, shapes, and patterns. Fine-tuning the model with regard to monument classification allows the network to utilize generic visual features for domain-specific characteristics, thus improving classification accuracy with little training data [2].

Furthermore, another noteworthy feature of the proposed method is the adoption of efficient preprocessing methods, particularly saliency-based cropping. Background information is an inevitable challenge in satellite images that may affect model

performance. Using saliency-based cropping in satellite images allows the model to extract relevant information mainly from the monument, reducing background interference and improving model accuracy and reliability [8], [9].

Finally, the proposed model solves several challenges encountered when dealing with satellite imagery analysis, including issues of scale, rotation, illumination, and resolution. Despite the above-listed challenges, the deep learning technique adopted in the proposed system enables the model to learn efficient and discriminative representations. As such, the system can successfully detect monuments with diverse conditions. Based on the experimental outcomes, the proposed system has proven effective and reliable, as it guarantees the classification accuracy rate of about 92.7%.

One of the main achievements of the proposed system is the implementation of various methods for explaining predictions of the machine learning model. In particular, Grad-CAM, LIME, and SHAP are used to make the decision-making process more comprehensible. Grad-CAM helps create intuitive heatmaps that reveal those areas of the image that contribute most to the decision making. With the help of LIME, local explanations can be generated by finding out the approximation of the behaviour of the model in the vicinity of the input sample. Finally, SHAP allows us to assign importance values to input features and generate explanations on local and global levels [3]–[5].

Thanks to the application of the XAI approach, the accuracy of the developed system increases significantly. By analyzing explanations, users can understand whether the model relies on relevant structural features of monuments (e.g., domes, arches, layouts), or on irrelevant background patterns.

Moreover, the suggested architecture shows stability during training and robust generalization properties. Similar results of training and validation suggest that the proposed solution is free of overfitting and can be deployed in real-world settings.

Nonetheless, there are several limitations associated with the proposed solution that should be addressed through further research. The effectiveness of the model relies on the quality and size of the dataset used for training and testing. For example, in cases when monuments are partly hidden, are depicted as tiny objects, or there are low-resolution images present in the dataset, the efficiency of the architecture may become lower. In

addition, despite being useful, the application of Explainable AI (XAI) techniques can increase computational costs and generate different explanations of the same prediction.

Future research may address these limitations by using larger datasets that would include various types of images, implementing advanced architectures of artificial neural networks, and improving XAI algorithms in terms of speed and reliability of explanations.

In summary, the IndiSight++ framework provides a highly effective and computationally efficient approach for recognizing monuments from satellite images. Through an effective integration of deep learning and XAI techniques, this study adds valuable contributions to building transparent and dependable intelligent systems for use in heritage conservation, remote sensing, and GIS applications.

5.2 Future Scope

Several possibilities for further research and advancement arise from the proposed IndiSight++ framework:

5.2.1 Increasing Datasets

Integrating more diverse and extensive datasets that include different types of monuments, various geographic locations, and differing environments will help strengthen the model's robustness and generalization. A broader dataset could assist the model in learning richer feature representations.

5.2.2 Higher Resolution Satellite Imagery

Using higher resolution satellite images will improve the model's capacity to detect subtle structural characteristics of the monuments. This could enhance classification accuracy when monuments occupy only a small section of the image or are partially covered.

5.2.3 Enhanced Architectures

Employing more advanced architectures like Efficient Net, Vision Transformers (ViTs), or hybrid CNN–transformer models can aid in improving feature extraction and

classification efficiency.

5.2.4 Real-time Implementation

Optimizing the model for real-time implementation on edge computing systems will facilitate practical applications, including real-time monitoring, surveillance, and automatic heritage mapping.

5.2.5 Integration with GIS Systems

Integration of the proposed method into GIS systems may help achieve better spatial analysis, mapping, and surveillance of monuments at the regional level.

5.2.6 Better Techniques for Enhancing Explainability

Efforts can be made in the future to develop more efficient techniques to improve explainability without increasing computational burden.

5.2.7 Multi-modal Recognition

Use of multiple modalities like text, history, and 3D reconstruction data may improve the effectiveness of monument recognition process.

A number of directions are indicated for further research and development based on the proposed IndiSight++ framework.

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Monument Recognition from Satellite Images Using Deep Learning with Explainable AI

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Abstract—Monument recognition using satellite images is a challenging problem for computer vision because of the variability in scale, complicated background, and lack of clarity about the structural features. In this paper, we propose a deep learning architecture named IndiSight++ for recognizing monuments using satellite images. We also used the concept of explainable AI for better interpretation and visualization of the results. In our proposed architecture, we used MobileNetV1 and satellite image preprocessing and augmentation strategies. To remove the background clutter, we also used saliency-guided cropping. We used Grad-CAM, LIME, and SHAP for visualization and explanation of our results. We collected some satellite images belonging to 10 different monument classes for experimentation. We achieved 92.7% accuracy on our test set. From the visualization results, we observed that our model is looking only at the structural features and not at the background.

Index Terms—Satellite Imagery, Monument Recognition, MobileNetV1, Explainable AI, Grad-CAM, LIME, SHAP

I. INTRODUCTION

India is known to have a rich heritage of architecture in the form of many historical monuments that signify a long history of cultural and architectural evolution. Such historical monuments can be helpful in various applications, such as heritage documentation, tourism and urban development analysis, and environmental monitoring. With recent developments in satellite imaging technology and the availability of satellite imagery from freely accessible sources, it is possible to identify historical monuments using satellite imaging.

Nevertheless, there are various problems associated with recognizing monuments through satellite images compared to



Fig. 1. Sample satellite collage of selected Indian monuments.

other landmark recognition methods via images taken from the ground level. For instance, there is a possibility that a monument may be viewed as a small part of a large image block and may be covered with other components such as vegetation, roads, water bodies, and even urban structures. Furthermore, satellite images may vary significantly with regard to scale, illumination, and orientation.

Convolutional neural networks [1] and other deep learning-based methods have proven to be very successful in various

computer vision and remote true to real cases. Nonetheless, it is not necessarily true that even when a model has a very good prediction capability, it is not necessarily true that it is recognizing any meaningful features of monuments. There is a possibility that it may be relying on trends from the background images that may be associated with other classes rather than the edifice of monuments. Such behavior makes an Android an object of trust in androids, especially with regard to various applications related to cultural heritage.

The concept of explainable Artificial Intelligence has been improved to make it easier to have transparency using insights into model decision-making processes. XAI methods can also be applied to ensure that the model is using the right visual evidence by figuring out which part of the image can be used to make the most important prediction. The importance of explainability is also significant in satellite image analysis since the context is very persuasive. Although XAI is gaining popularity in remote sensing, it is still underutilized as part of the pipeline in full monument recognition systems, which is an area where we hope to contribute.

In this paper, the framework is presented and is referred to as IndiSight++. It is satellite-based and is capable of identifying monuments in images. Emphasis is placed on classification and interpretation. In the proposed model, there is the use of a lightweight classifier referred to as the mobile net V1 classifier. There is also the use of saliency cropping for minimizing interference from the background, which is in line with the efficiency aspect of the student projects. Various techniques for explaining the model's output in the form of Grad-CAM, LIME, and SHAP have also been integrated into the model.

From the experimental results, it is clear that the framework is capable of recognizing monuments and is interpretable. Sample satellite images used in the dataset are presented in Fig. 1.

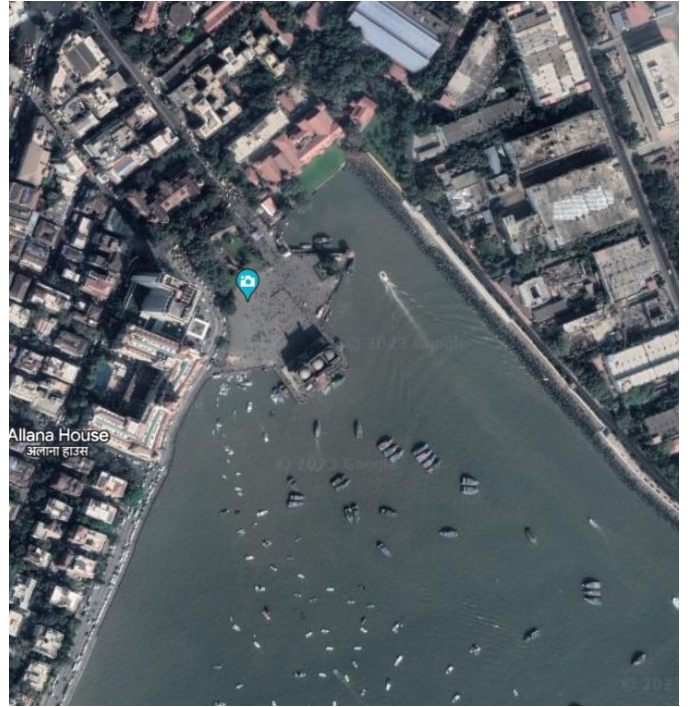
Deep learning models, particularly convolutional neural networks (CNNs), have significantly improved performance in various computer vision tasks including image classification, object detection, and scene understanding. Architectures such as MobileNet and ResNet have demonstrated strong capability in extracting hierarchical visual features while maintaining computational efficiency for large-scale image analysis tasks [1], [2]. These models have also been successfully applied in remote sensing image interpretation and land-use classification [13], [17].

The main contributions of this work are:

- A lightweight satellite-based monument recognition framework using MobileNetV1.
- A saliency-guided crop proposal strategy to mitigate background dominance.
- Integration of Grad-CAM, LIME, and SHAP for complementary explainability.
- A curated dataset of some satellite images across 10 monument classes.
- Deployment insights for web-based and edge inference.



(a) Taj Mahal



(b) Gateway of India

Fig. 2. Representative satellite tiles from the dataset.

II. RELATED WORK

Monument and landmark recognition at ground level with ground level images has been mostly researched. The initial ones were founded on artisanship characteristics including SIFT and SURF, which is normally matched with bag-of-visual-words models and geometrical verification. The boom of deep learning has been substituted by the convolutional neural networks, which have enabled the landmark recognition to

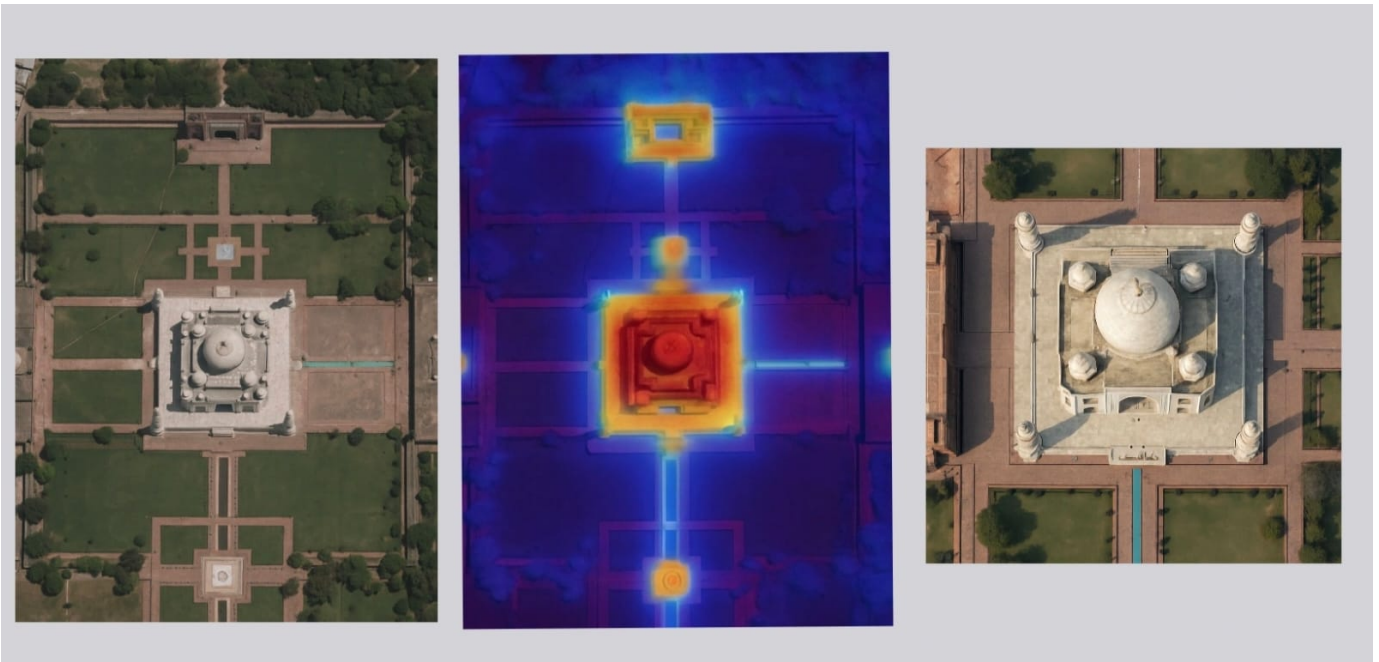


Fig. 3. Saliency map generation and crop proposal process.

be conducted on a large scale and photo-selected geolocation of a street-level shot.

On the other hand, satellite and aerial image analysis has been a common practice throughout the science of remote sensing in the categorizing of scenes, mapping of land-use and object detection. Deep learning systems are found to be good in detecting spatial patterns in overhead images. However, most of the studies refer to the broad categories of the scenes or large objects and up to now, there is still relative defocus on fine-grained monument recognition, specifically, when the coverage area of the object is restricted to a small locality within the frame of the crowded urban set-up.

Papers that have dealt with object or landmark recognition in aerial images have configured it to object or multi-class recognition. Many of these techniques provide a fairly good level of accuracy, they are more performance oriented and they do not provide much information on the manner in which the model is making its decisions. In satellite-based monument recognition, non-interpretability is a problem, as local contextual features may corrupt the predictions. One trend that has become extremely popular in ensuring that deep learning model is a more comprehensible one is explainable AI (XAI). They include such methods as Grad-CAM that offer visual highlight of the discriminative areas, and concepts such as LIME and SHAP that interpret decision on feature-level based on perturbations. Some of the activities that have been performed in remote sensing with XAI include land-cover classification and change-detection, with the primary reasons being to build the trust of a user and to make the model credible.

In the recent theme, there are some studies which have

focused on the more adaptive classification dotted such as random vector functional link (RVFL) networks and extensions based on the intuitionistic fuzzy added to develop a sense of robustness and generalization in more intricate versions of classification [18]–[20]. The indications of success by these innovative techniques are that hybrid procedures between deep feature segmentation and adaptive classifiers can achieve success and lead the discourse to more interpretable and lightweight approaches to a vision task.

Despite such developments, most of the studies include the use of them as post-hoc tools, but not as components of the entire process of inference. Also, they do little work with numerous XAI techniques in investigating model behavior of overhead images. The specified paper will aim to cover these lapses by incorporating a hybrid of multiple XAI techniques into a single satellite-based monster recognition system and discussing how the explanations will be connected with the context-based visual clues on the one hand and the one associated with the monument on the other hand.

Convolutional neural networks have been utilized to conduct large-scale landmark recognition as geolocation and visual place recognition. Such strategies acquire discriminative visual patterns that can be used to detect a landmark in varying viewpoints and environments. [14].

The catalog of data and benchmarks in remote sensing has been significant in promoting the study of object detection with respect to aerial images. The DOTA dataset is one such example which offers a large-scale benchmark to detect objects in aerial images to develop powerful deep learning models to analyze overhead images [16].

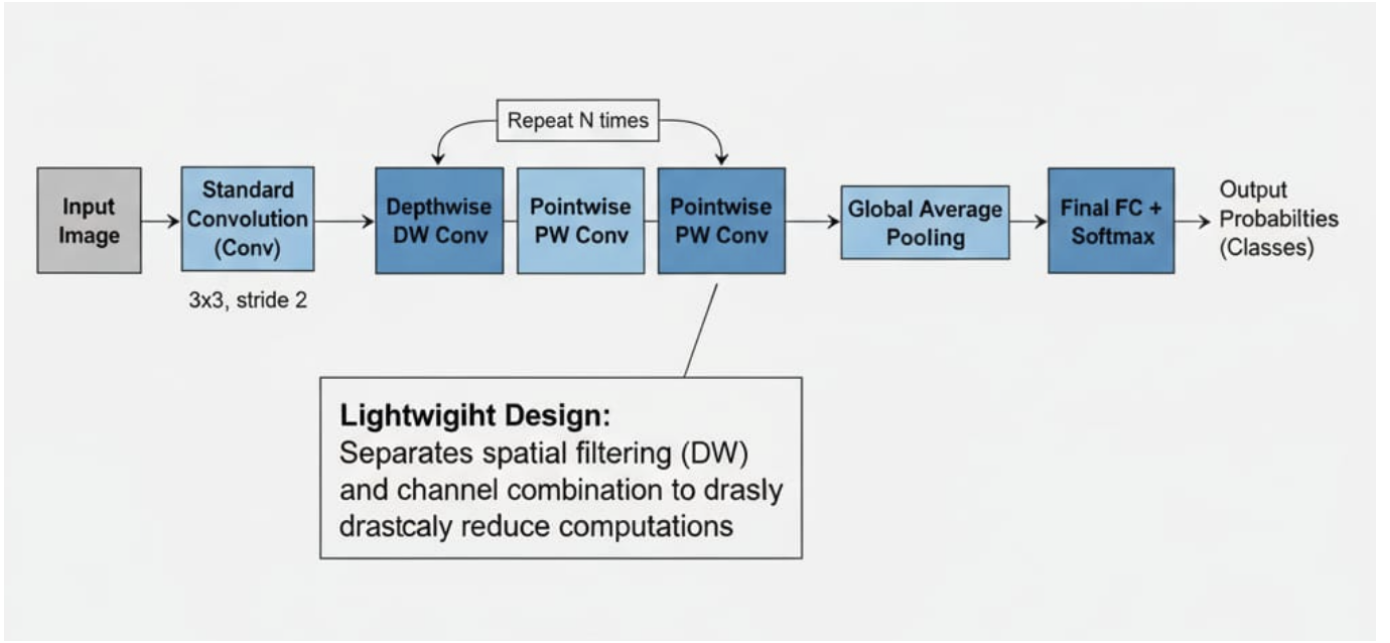


Fig. 4. Overview of the MobileNetV1 architecture.

III. DATASET AND ANNOTATION

Satellite images used in this study were collected from publicly available geospatial data services like Google Earth, Bing Maps, and Bhuvan. These services provide satellite images of good quality that can be used for research purposes in remote sensing and geospatial technologies [8]–[10]. Ten major Indian monuments were considered for this study, and a some satellite images were collected. Images with cloud presence or incorrect image labels were excluded from the dataset. Fig. 2 illustrates examples of satellite images from the dataset.

IV. PREPROCESSING AND SALIENCY PROPOSAL

Data preprocessing is one of steps in satellite image preparation to deep learning models. Raw satellite images usually have a variation in resolution, lights, and atmospheric noise which can have adverse implications on the performance of the model. Thus, the quality and consistency of the dataset were enhanced with the help of preprocessing methods.

The satellite visuals employed in the research were gathered on publicly accessible geo-spatial sites such as Google Earth, Bing Maps and Bhuvan [8]–[10]. Manual inspection was done to remove images with cloud cover, occlusions and incorrect labels to maintain quality of data sets.

After that, the resizing of all images to a similar resolution used in the deep learning model was done. The intensity values were scaled to a standard range by using pixel normalization which is utilized to enhance the stability of the training. Besides this, data augmentation methods which included rotation and horizontal flipping were used to make data more diverse and minimize the likelihood of overfitting.

They also used the saliency detection methods to find the areas that are the most visually prominent in the satellite images [6]. These methods emphasize significant regions by assessing the difference between the local structures and the surrounding environment. Spectral residual method offers a computationally inexpensive technique of producing saliency maps which highlight possible object areas in complex images. Fig. 2 displays representative examples of the processed satellite images.

Lastly, the processed images were sorted based on their respective classes of monuments and categorized into training and evaluation sets in the form of training and validation.

V. MODEL ARCHITECTURE AND TRAINING

MobileNetv1 is proposed in the present model because it is a very efficient design that has little computational demands. MobileNetV1 uses depthwise separable convolutions, which have lower parameters and less computational complexity than conventional convolutional neural networks, but still have good classification ability. This renders it appropriate to the satellite image classification.

VI. RESULTS AND DISCUSSION

The model was trained with the weights that have been pre-trained on the ImageNet dataset through the transfer learning strategy. Using pre-trained weights enables the network to use visual features that have been learned in the past like an edge and a text featuring making the training faster and more accurate.

The last fully connected layer was adjusted to the size of the number of classes in the dataset of monuments to make the architecture adapt to the classification of monuments. The model was also trained on the Adam optimizer and

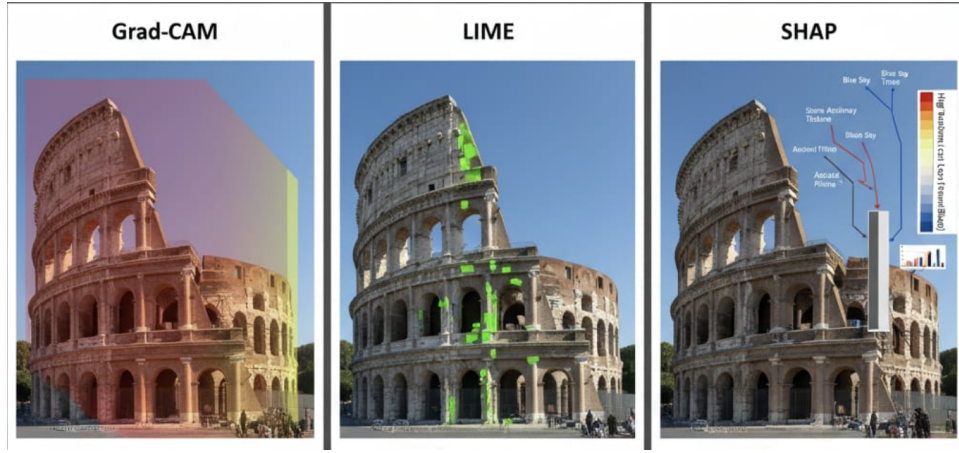


Fig. 5. Combined explainability outputs using Grad-CAM, LIME, and SHAP.

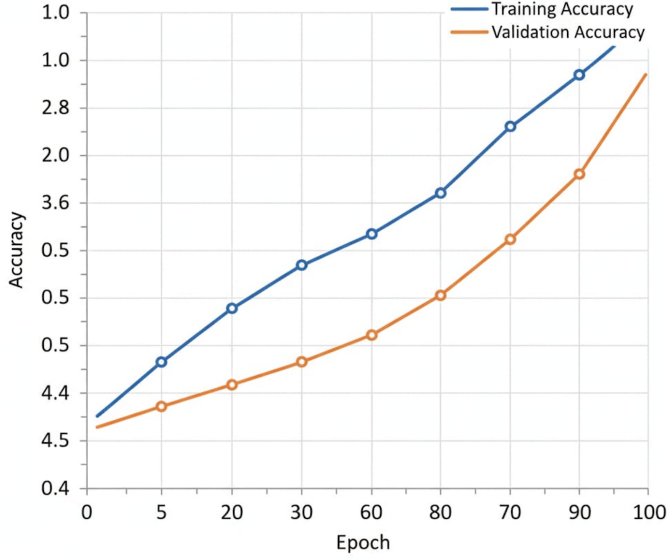


Fig. 6. Training and validation accuracy curves.

Fig. 7. Web-based interface displaying predictions and explanations.

categorical cross-entropy loss that is often used on multi-class classification. The dataset has been broken down into training, validation and testing sets in order to test the performance of the model and avoid overfitting.

A general design of MobileNetV1 architecture applied in this paper is shown in Fig. 4.

VII. EXPLAINABILITY METHODS

Grad-CAM finds the key spatial locations, LIME finds those that are key locally and SHAP finds additive scores to features that are key locally.

To enhance the level of transparency and interpretability of a deep learning model, explainable Artificial Intelligence (XAI) techniques have been proposed [21]. Grad-CAM and other gradient-based visualization techniques identify discriminative functions of a model in grad values, which are used in a geometric location highlighted on a map [4]. Model-agnostic methods such as LIME and SHAP give local explanations by examining contribution of features and perturbations in the input data to the model [3], [5]. These methods are also becoming a common method of remote sensing to gain a deeper insight into model behavior [15].

The explainability results combined in Fig. 5 in this paper represent the areas on the satellite images that the model relied on the most to make its predictions.

The system proposed was able to attain an accuracy of around 92.7% in classification. The training process showed convergence, which means it was able to learn features from the satellite images.

Transfer learning using MobileNetV1 was able to enhance the model's performance.

Fig. 6 shows the training and validation accuracy curves. These curves show the learning curve of the model during training. The curves show consistency in improving accuracy with little variation between training and validation accuracy, which means it did not overfit.

Furthermore, the explanation maps created with the aid of gradient-based visualization methods clearly indicate the

model's emphasis on the structural parts of the monuments, as opposed to the background. This, therefore, suggests that the model has effectively learned significant spatial features related to the shapes and configurations of the monuments, with minimal concern for the context.

VIII. DEPLOYMENT CONSIDERATIONS

To enable the visualization of the predictions and their explanations in real-time during the course of the project work, a web interface using Flask was developed. This model can be further converted to TensorFlow Lite, enabling deployment to edge devices, which can be utilized for coursework. New-age AI-aided development platforms such as ChatGPT and Gemini are being employed for research activities, including experimentation, documentation, and rapid prototyping of ML systems [11], [12].

IX. CONCLUSION

In this paper, we discuss IndiSight++, an Indian monuments recognition system that makes use of automated recognition of Indian monuments using satellite imagery controlled by integrating deep learning and explainable AI. It resolves problems of large variations in scale, background clutter, and low resolutions in satellite imagery with a small MobileNet V1 and saliency preprocessing.

The model was tested against a filtered set of satellite images with ten categories of monuments and showed promising classification results with being interpretable. Grad-CAM, LIME, and SHAP were used to present alternative insights of the decision-making process of the model. These methods of explainability assist in determining whether the model uses structural characteristics of monuments as a base of its predictions or simply used another contextual factor. This interpretability is needed where cultural heritage analysis applications and decision support systems are involved and the reliability and transparency are vital.

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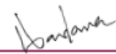
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